



AMRITA
VISHWA VIDYAPEETHAM

USER INTERFACE DESIGN LAB MANUAL

SUBMITTED BY:

NAME	PURAMSHETTY SRISHANTH
ROLL NO	AV.SC.U4CSE25234
YEAR/SEM	1 ST YEAR/SEM-2
CREDITS	3
SECTION	CSE-C

SUBMITTED TO:

NAME	DR.RAJKUMAR BATCHU SIR
DESIGNATION	Assistant Professor (Sl. Gd.)
DEPARTMENT	School Of Computing

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	a button using html and css B) Navigation using iframe (mini website) upon clicking each buttons the details about that button should be display on the same page. C) Design amrita login page by showcampus on home page.upon clicking on each campus the details about each campus should be display on the same page use html and css to design a stylish webpage			
6.	Week 6:A) Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances. B) Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances, books. Based on the category we select corresponding products should be displayed on the sidebar.	62-72		

7.	Week 7: create a GitHub account and create a repository in it and push all the week 6 code files in it.	73-75		
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9.	Week 9:Create a webpage using html,css,js based on the give conditions: A:create a button for each program , based on the button clicked the corresponding program has to be executed and should display the output. B:the logics for each button includes finding an even or odd number , prime or not , factorial of a number , Armstrong or not. C:use appropriate design to display the output	83-88		
10.	Week 10:A:write JavaScript conditional statement to find the sign of product of three numbers using functions .Display an alert box with specified sign.	89-94		

	B:write a java script conditional statements to sort 3 numbers using functions display an alert box to show the result. C:write a java script conditional statement to find largest of five numbers using function and onclick event .display an alert box to show the result D:write a java script for loop that will iterate from 0 to 15. For each iteration ,it will check if the current number odd or even , and display a message on screen E:write a java script program to read 6 subjects marks from the student then computes the average marks of students then , this average is used to determine the corresponding grade.			
11	11a) A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters. 11b) A registration form should validate email format before submission. If invalid, the form should not be submitted. 11c) A system should check password strength while typing and display whether it is weak or strong. 11d) A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits. 11e) Design an HTML page, such that the registration form must validate:	95-106		

12.	Installation of electron	107-118	
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Week1:

1. Installation of VS Code tool.
2. Understanding basics of html stucture.
3. Understanding basic html tags.

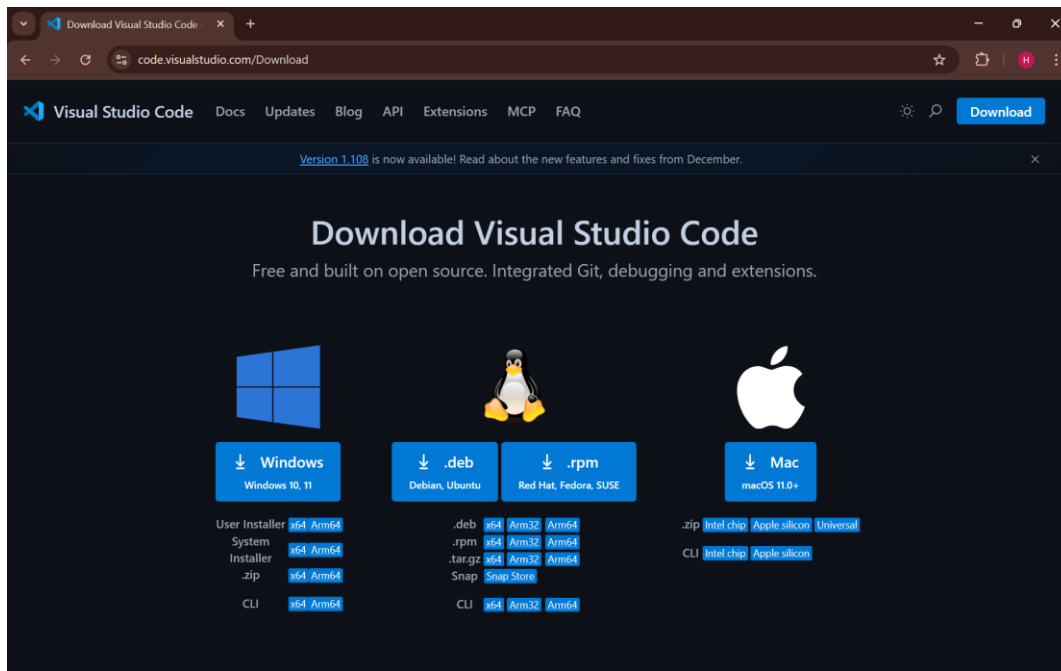
1a) AIM: Installation of VS code tool

Procedure:

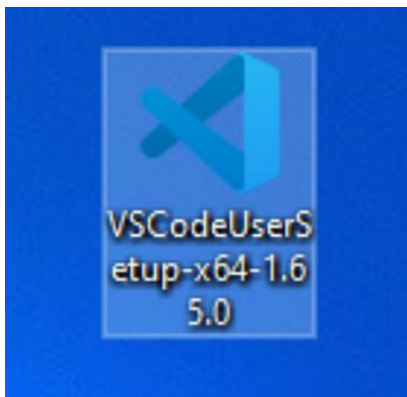
Step 1: Visit the Official Website of the Visual Studio Code using any web browser like [Google Chrome](#), [Microsoft Edge](#), etc.



[Step 2]: Press the "Download for Windows" button on the website to start the download of the Visual Studio Code Application.

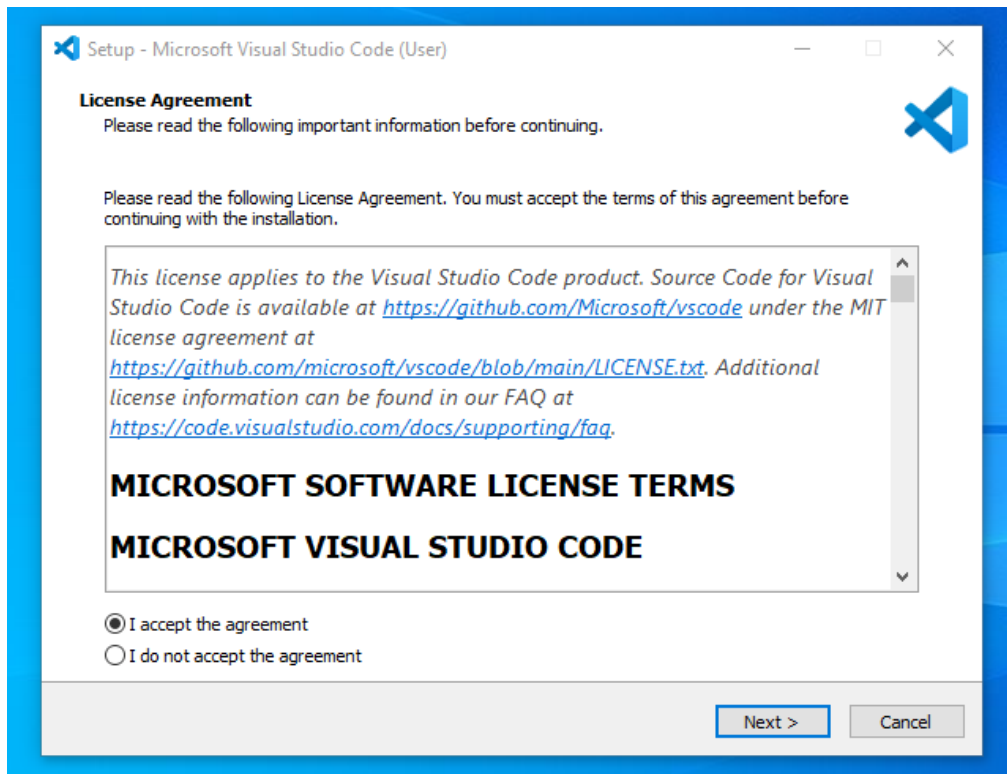


[Step3] When the download finishes, then the Visual Studio Code Icon appears in the downloads folder.

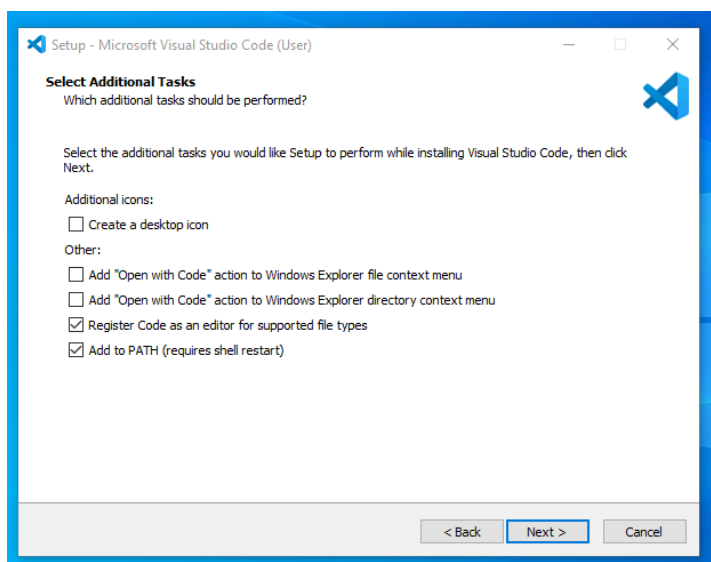


[Step4] Click on the Installer icon to start the installation process of the Visual Studio Code.

[Step5] After the Installer opens, it will ask you to accept the terms and conditions of the Visual Studio Code. Click on I accept the agreement and then click the Next button.

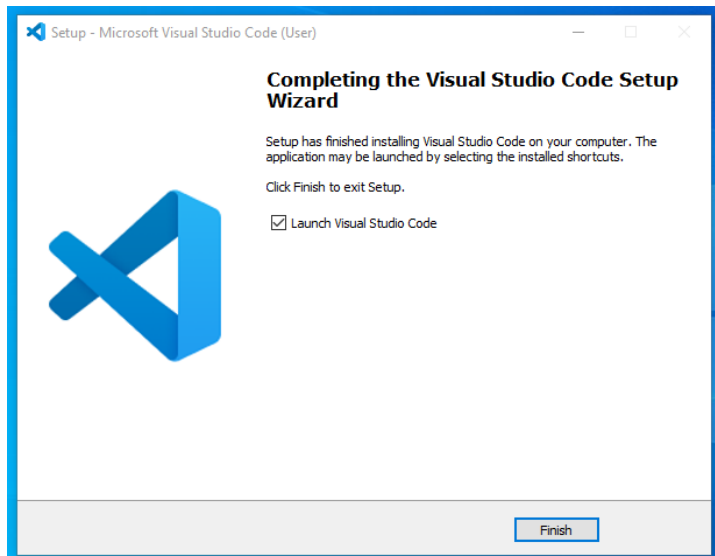


[Step6]:Choose the location data for running the Visual Studio Code. It will then ask you to browse the location. Then click on the Next button.

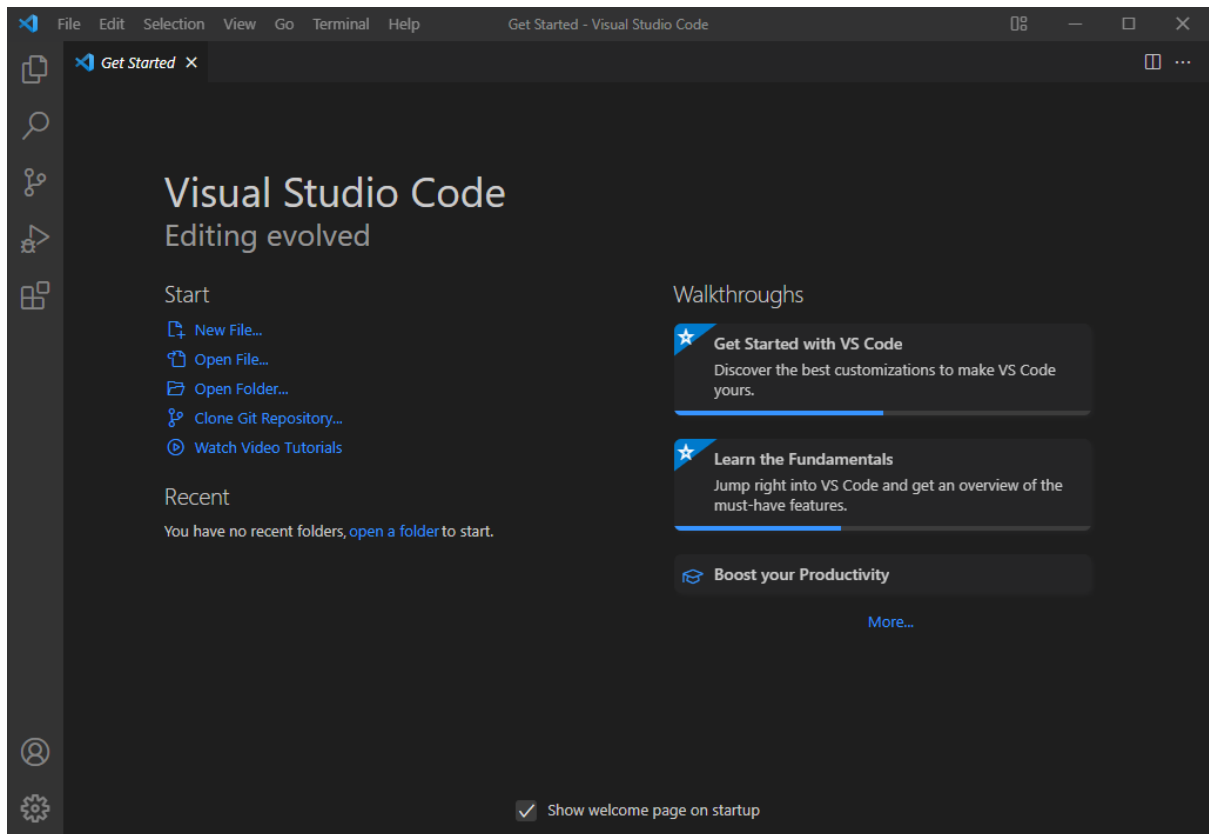


[Step7]Then it will ask to begin the installation setup. Click on the Install button. After clicking on Install, it will take about 1 minute to install the Visual Studio Code on your device.

After the Installation setup for Visual Studio Code is finished, it will show a window like this below. Tick the "Launch Visual Studio Code" checkbox and then click Next.



[Step 10]: After the previous step, the Visual Studio Code window opens successfully. Now you can create a new file in the Visual Studio Code window and choose a language of yours to begin your programming journey!



So this is how we successfully installed Visual Studio Code on our Windows system. The steps mentioned in the above guideline can be used in any kind of Windows Browsers for Downloading & Installation VS Code on Windows 10. So, try these steps to get the VS Code IDE on your device as well.

1b). Aim: To Understand the Structure of HTML

Procedure:

Puramshetty srishanth;AV.SC.U4CSE25234;CSE C

```
<!DOCTYPE html>
```

```
<html>
```

```
<title>
```

```
    HTML BASICS
```

```
</title>
```

```
<body>
```

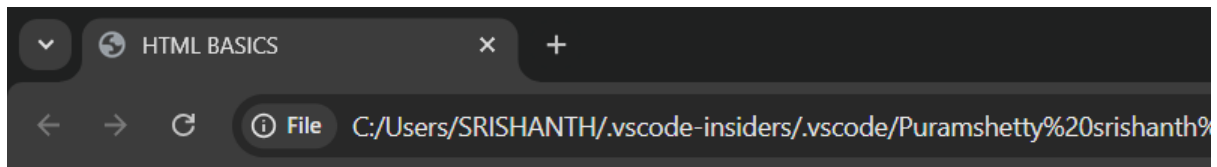
```
    <h1>FIRST HEADING</h1>
```

```
    <p>FIRST PARAGRAPH</p>
```

```
</body>
```

```
</html>
```

Output:



Puramshetty srishanth;AV.SC.U4CSE25234;CSE C

FIRST HEADING

FIRST PARAGRAPH

EXPLANATION:

- HTML: HYPER TEXT MARKUP LANGUAGE
- Used To Create And Structure Webpages
- <!DOCTYPE HTML>: tells that the doc type is html5
- <html>: The element is the root element of an HTML page.
- <head>: The element contains meta information about the HTML page
- <title>: The <title> element specifies a title for the HTML page.
- <body>: The <body> element contains all visible content like text images, link, tables, etc.
- <h1>: the <h1> element defines a large heading
- <p>: The <p> element defines a paragraph

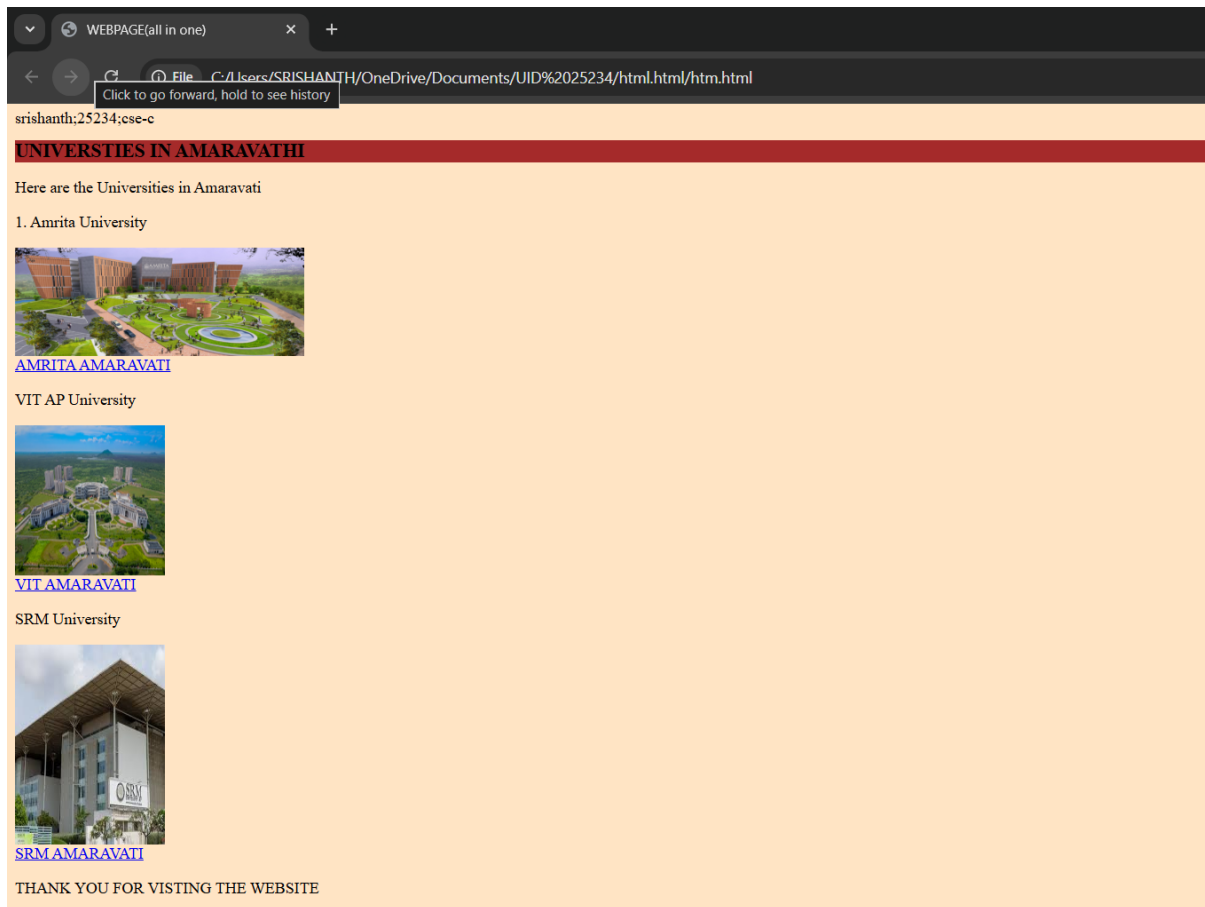
1c) Aim : understanding basic html tags.

Tags - <p>; <h1>; ;<color>

CODE:

```
html.html > > html.html > html > head > title
1  srishanth;25234;cse-c
2  <!DOCTYPE HTML>
3  <html>
4  <head>
5  <title>
6  |  WEBPAGE(all in one)
7  </title>
8  </head>
9  <body style="background-color: #bisque;">
10 <h1 style="background-color: #brown; font-size: larger;">
11 |  UNIVERSITIES IN AMARAVATHI
12 </h1>
13 <p>
14 |  Here are the Universities in Amaravati
15 </p>
16 <p>
17 |  1. Amrita University
18 </p>
19 
20 <br>
21 <a href="https://www.amrita.edu/campus/amaravati/">AMRITA AMARAVATI</a>
22 <p>
23 |  VIT AP University
24 </p>
25 
26 <br>
27 <a href="https://vitap.ac.in/">VIT AMARAVATI</a>
28
29 <p>
30 |  SRM University
31 </p>
32 
33 <br>
34 <a href="https://www.srmap.edu.in/">SRM AMARAVATI</a>
35 <p>
36 |  THANK YOU FOR VISTING THE WEBSITE
37 </p>
38 </body>
39 </html>
40
```

OUTPUT:



Explanation:

- `<!DOCTYPE HTML>`: tells that the doc type is html5
- `<html>`: The element is the root element of an HTML page.
- `<head>`: The element contains meta information about the HTML page
- `<title>`: The `<title>` element specifies a title for the HTML page.
- `<body>`: The `<body>` element contains all visible content like text images, link, tables, etc.
- `<h1>`: the `<h1>` element defines a large heading

- `<p>`: The `<p>` element defines a paragraph
- `<br>`: it is used to break the lines
- `<img src="">`: it is used to add image
- `<a href="" >`: it is used to add link in the html website

Week-2

2a) Aim: Design a basic webpage with headings , paragraph, pre, table by adding styles.

```
<!DOCTYPE html>
<html>
  <head>
    <title>website</title>
  </head>

  <body style="background-color:rgb(231, 4, 4);background-
image:url('amrita.jpg');">
    <h1 style="text-align:center;color:crimson;border: 7px solid rgb(238,
155, 130);">AMRITA VISHWA VIDYAPEETHAM</h1>

    <pre style="text-align: center;font-style:italic;color:rgb(152, 25,
25);font-size:x-large;">
      Amrita Vishwa Vidyapeetham is a leading multi-disciplinary and
research-intensive

      university, establishing its campus in Amaravati. Nestled amidst
the scenic beauty of the

      Krishna River and surrounded by natural wonders and historical
sites, Amaravati city

      embodies the principles of sustainability and liveability. We are
building one of the most

      buoyant centre of the research, education and technological
advancement, in the all new

      greenfield capital city.

    <table border="1" cellpadding="5" cellspacing="10" width="60%"
align="center" bgcolor="beige" style="color:rgb(0, 0, 0);font-family: Georgia,
'Times New Roman', Times, serif;">
      <tr style="background-color: aqua;">
        <th colspan="6">AMRITA VISHWA VIDHYAPEETHAM BRANCHES</th>
      </tr>

      <tr style="background-color: rgb(253, 182, 89);color: rgb(14,
42, 183);">
```

```

        <th>S.No</th>
        <th>Campus</th>
        <th>Location</th>
        <th>Contact No.</th>
        <th colspan="2">Contact Person</th>
    </tr>

    <tr style="background-color: rgb(253, 182, 89);color: rgb(14,
42, 183);">
        <td>1</td>
        <td>Amaravati Campus</td>
        <td>Kuragallu</td>
        <td>9874561230</td>
        <td>leela</td>
        <td>sravanthi</td>
    </tr>

    <tr style="background-color: rgb(253, 182, 89);color: rgb(14,
42, 183);">
        <td>2</td>
        <td>coimbatore</td>
        <td>Marudamalai</td>
        <td>9758457898</td>
        <td colspan="2">sachin singh</td>

    </tr>

    <tr style="background-color: rgb(253, 182, 89);color: rgb(14,
42, 183);">
        <td>3</td>
        <td>Chennai Campus</td>
        <td>Basic electronic</td>
        <td>6987458545</td>
        <td>Akash</td>
        <td>sunny</td>
    </tr>
</table>
</pre>
</body>
</html>

```

OUTPUT



AMRITA VISHWA VIDYAPEETHAM

Amrita Vishwa Vidyapeetham is a leading multi-disciplinary and research-intensive university, pioneering the knowledge frontier, inspired by the ancient wisdom of the Krishna River and surrounded by natural wonders and historical sites, Amaravati city embodies the principles of sustainability and livability. We are building one of the best student centers of the research, education and technological advancement, in the all time pursuit of excellence.

AMRITA VISHWA VIDHYAPEETHAM BRANCHES				
S.No	Campus	Location	Contact No.	Contact Person
1	Amaravati Campus	Kuragallu	9874561230	leela sravanthi
2	coimbatore	Marudamalai	9758457898	sachin singh
3	Chennai Campus	Basic electronic	6987458545	Akash sunny

WEEK-3

1a) Create a class time table using a HTML elements.

```
<html>
  <head>
    <title> timetable</title>
  </head>
  <table border="3">
    <caption style="background-color: yellow;">SEM-2 TIMETABLE</caption>
    <tr>
      <th>DEPT.CSE </th>
      <th rowspan="3" colspan="10">ROOM VENUE 512</th>
    </tr>
    <tr>
      <th>CLASS BTECH CSE-C</th>
    </tr>
    <tr>
      <th>SEMESTER-2</th>
    </tr>
    <th style="font-family: 'Lucida Sans', 'Lucida Sans Regular',
'Lucida Grande', 'Lucida Sans Unicode', Geneva, Verdana, sans-serif;"
colspan="3"> Advisors:
      Name: Mr. Budati Jaya Lakshminarayana
      Email : bj_lakshminarayana@av.amrita.edu
      Mobile Number :+91-7989795768 </th>

      <th style = "font-family: 'Lucida Sans', 'Lucida Sans Regular',
'Lucida Grande', 'Lucida Sans Unicode', Geneva, Verdana, sans-serif;"="8">
Name: Dr. Arun Kumar S
      Email : s_arunkumar@av.amrita.edu
      Mobile Number : +91-8940857202</th>
    </tr>
  </table>
  <table border="3" style="border-collapse: collapse;">
    <tr>
```

```

        <th rowspan="2">TIME TABLE</th>
        <th>SLOT-1</th>
        <th>SLOT-2</th>
        <th>SLOT-3</th>
        <th>SLOT-4</th>
        <th>SLOT-5</th>
        <th>SLOT-6</th>
        <th>SLOT-7</th>
        <th>SLOT-8</th>
        <th>SLOT-9</th>
        <th>SLOT-10</th>

    </tr>
    <tr>
        <th>8:50-9:40</th>
        <th>9:40-10:30</th>
        <th>10:30-10:40</th>
        <th>10:40-11:30</th>
        <th>11:30-12:20</th>
        <th>12:20-1:10</th>
        <th>1:10-2:30</th>
        <th>2:30-3:20</th>
        <th>3:20-4:10</th>
        <th>4:10-5:00</th>

    </tr>
    <tr>
        <td>MONDAY</td>
        <td colspan="2">23MAT116 Discrete Mathematics_lab2</td>
        <td rowspan="5" style="background-color: yellow; writing-mode:
vertical-lr; text-align: center ; ">short break</td>
        <td>23PHY115 Modern physics</td>
        <td>23CSE111Object
Oriented Programming</td>

        <td >library</td>
        <td rowspan="5" style="background-color: yellow;writing-mode:
vertical-lr; text-align: center;"> lunch</td>
        <td>library</td>
        <td rowspan="5" style="background-color: yellow; writing-mode:
vertical-lr; text-align: center;">short break</td>

        <td colspan="2">Library/Sports</td>
    </tr>
        <td>TUESDAY</td>
        <td colspan="2">23CSE113 User Interface Design LAB(BYOD)</td>

```

```

        <td>23MAT117 Linear
Algebra</td>
        <td>23MAT116 Discrete
Mathematics</td>
        <td COUNSELLING style="color: red;">COUNSELLING</td>

        <td>library </td>

        <td colspan="2">22ADM111 Glimpses of Glorious India</td>

    </tr>
    <tr>
        <td>WEDNESDAY</td>
        <td> 23PHY115 Modern
Physics</td>
        <td>23CSE113 User Interface
Design</td>

        <td colspan="2">23MAT117 Linear Algebra_LAB(BYOD)</td>
        <td COUNSELLING style="color: red;"> COUNSELLING</td>

        <td>library</td>

        <td colspan="2">Library/Sports</td>
    </tr>
    <tr>
        <td>THURSDAY</td>
        <td>23CSE1110bject
Oriented
Programming</td>
        <td>23MAT116 Discrete
Mathematics</td>

        <td colspan="2">23CSE1110bject Oriented Programming
_Lab2</td>
        <td>23MAT117 Linear
Algebra</td>

        <td>23MEE115 Manufacturing
Practice LAB</td>

        <td colspan="2">23MEE115 Manufacturing
Practice LAB</td>
        <td></td>

```

```

        </tr>
        <tr>
            <td>FRIDAY</td>
            <td>23CSE1110bject
Oriented
Programming</td>
            <td>23MAT116 Discrete
Mathematics</td>

            <td>23PHY115 Modern
Physics(T)</td>
            <td>23CSE113 User
Interface Design</td>
            <td>23MAT117 Linear
Algebra</td>

            <td>22ADM111Glimpses of
Glorious India(P)</td>

            <td colspan="2">Library/Sports</td>
        </tr>

    </tr>
    <tr>
        <br>

    </tr>
</table>
<br>

<table border="3" width="100%">

    <tr>
        <th>course code</th>
        <th>course name</th>
        <th>L-T-P</th>
        <th>credits</th>
        <th>faculty name</th>
        <th>assisting faculty</th>

    </tr>
    <tr>
        <td>23MAT116</td>

```

```

        <td>Discrete Mathematics</td>
        <td> 3-0-2 </td>
        <td>4</td>
        <td> Dr.Srivivasa Rao Thota</td>
        <td></td>

</tr>
<tr>
        <td>23MAT117</td>
        <td>Linear Algebra</td>
        <td>3-0-2</td>
        <td>4</td>
        <td>Mr.Sapavat Bixapathi</td>
        <td></td>

</tr>
<tr>
        <td>23CSE111</td>
        <td>Object Oriented Programmin</td>
        <td>3-0-2</td>
        <td>4</td>
        <td>Dr.SATYA RAJENDRA SINGH R</td>
        <td> Dr. Balaji TK</td>
</tr>
<tr>
        <td>23PHY115</td>
        <td>Modern Physics</td>
        <td>2-1-0</td>
        <td>3</td>
        <td> Dr.Soumya</td>
        <td></td>
</tr>
<tr>
        <td>23CSE113</td>
        <td>User Interface Design</td>
        <td>2-0-2</td>
        <td>3</td>
        <td> Dr.Rajkumar Batchu</td>
        <td> Mr. Barge Bharadwaj</td>
</tr>
<tr>
        <td>23MEE115</td>
        <td>Manufacturing practice</td>

        <td>0-0-3</td>
        <td>1</td>
        <td>Dr. Sathies S.</td>

```



```

        <td></td>
    </tr>
    <tr>
        <td>22ADM111</td>
        <td>Glimpses Of Glorious India</td>
        <td>2-0-1</td>
        <td>2</td>
        <td> Dr. Siva Panuganti</td>
        <td></td>

    </tr>
    <tr>
        <td></td>
        <td>Total (15L + 1Tut +12P =28 hrs)</td>
        <td>15-1-12</td>
        <td>21</td>
        <td></td>

        <td></td>
    </tr>

</table>
</html>

```

OUTPUT:

SEM-2 TIMETABLE										
DEPT.CSE						ROOM VENUE 512				
CLASS BTECH CSE-C										
SEMESTER-2										
Advisors: Name: Mr. Budati Jaya Lakshminarayana Email : bj_lakshminarayana@av.amrita.edu Mobile Number :+91-7989795768						Name: Dr. Arun Kumar S Email : s_arunkumar@av.amrita.edu Mobile Number : +91-8940857202				
TIME TABLE	SLOT-1 8:50-9:40	SLOT-2 9:40-10:30	SLOT-3 10:30-10:40	SLOT-4 10:40-11:30	SLOT-5 11:30-12:20	SLOT-6 12:20-1:10	SLOT-7 1:10-2:30	SLOT-8 2:30-3:20	SLOT-9 3:20-4:10	SLOT-10 4:10-5:00
MONDAY	23MAT116 Discrete Mathematics_lab2		short break	23PHY115 Modern physics	23CSE111 Object Oriented Programming	library	lunch	library	short break	Library/Sports
TUESDAY	23CSE113 User Interface Design LAB(BYOD)			23MAT117 Linear Algebra	23MAT116 Discrete Mathematics	COUNSELLING		library		22ADM111 Glimpses of Glorious India
WEDNESDAY	23PHY115 Modern Physics	23CSE113 User Interface Design		23MAT117 Linear Algebra_LAB(BYOD)		COUNSELLING		library		Library/Sports
THURSDAY	23CSE111 Object Oriented Programming	23MAT116 Discrete Mathematics		23CSE111 Object Oriented Programming_Lab2		23MAT117 Linear Algebra		23MEE115 Manufacturing Practice LAB		23MEE115 Manufacturing Practice LAB
FRIDAY	23CSE111 Object Oriented Programming	23MAT116 Discrete Mathematics		23PHY115 Modern Physics(T)	23CSE113 User Interface Design	23MAT117 Linear Algebra		22ADM111 Glimpses of Glorious India(P)		Library/Sports
course code	course name			L-T-P	credits	faculty name		assisting faculty		
23MAT116	Discrete Mathematics			3-0-2	4	Dr.Srivivasa Rao Thota				
23MAT117	Linear Algebra			3-0-2	4	Mr.Sapavat Bixapathi				
23CSE111	Object Oriented Programmin			3-0-2	4	Dr.SATYA RAJENDRA SINGH R		Dr. Balaji TK		
23PHY115	Modern Physics			2-1-0	3	Dr.Soumya				
23CSE113	User Interface Design			2-0-2	3	Dr.Rajkumar Batchu		Mr. Barge Bharadwaj		
23MEE115	Manufacturing practice			0-0-3	1	Dr. Sathies S.				
22ADM111	Glimpses Of Glorious India			2-0-1	2	Dr. Siva Panuganti				
	Total (15L + 1Tut +12P =28 hrs)			15-1-12	21					

2b)create an html web page that would display syllabus of all the subjects as an image format also that contain time table

```
Name = Srishanth , Roll no. = 25234 , Section-C
<!DOCTYPE html>
<html>
  <head>
    <title>Time table</title>
  </head>
  <body style="background-color: rgb(255, 255, 255);">
    <table width="100%" border="1" cellpadding="1" cellspacing="2"
style="text-align: center; background-color: rgb(255, 255, 255);">
      <tr>
        <th colspan="12">Master Time Table CSE II Semester C
Section</th>
      </tr>
      <tr>
        <td colspan="4" align="left"><b>Dept. CSE</b><br><b>Class: B.
Tech. CSE-C</b><br><b>Semester-II</b></td>
        <td colspan="8" align="left"><b>Class Room Venue: 512</b></td>
      </tr>
      <tr>
        <th colspan="4" align="left"><b>class Advisors:<br>1. Name:
Mr. Budati Jaya Lakshminarayana<br>Email :
bj_lakshminarayana@av.amrita.edu<br>Mobile Number :+91-7989795768</b></th>
        <th colspan="8" align="left"><b>2. Name: Dr. Arun Kumar
s<br>Email : s_arunkumar@av.amrita.edu<br>Mobile Number : +91-
8940857202</b></th>
      </tr>
      <tr>
        <th rowspan="2">Time/Day</th>
        <th><b>Slot-1</b></th>
        <th><b>Slot-2</b></th>
        <th rowspan="2">10:35AM - 10:45Am</th>
        <th><b>Slot-3</b></th>

```

```

        <th><b>Slot-4</b></th>
        <th><b>Slot-5</b></th>
        <th rowspan="2"><b>1:15PM - 2:10PM</b></th>
        <th><b>Slot-6</b></th>
        <th rowspan="2"><b>3:00PM - 3:10PM</b></th>
        <th><b>Slot-7</b></th>
        <th><b>Slot-8</b></th>
    </tr>
    <tr>
        <th><b>8:55AM - 9:45AM</b></th>
        <th><b>9:45AM - 10:35AM</b></th>
        <th><b>10:45AM - 11:35AM</b></th>
        <th><b>11:35AM - 12:25PM</b></th>
        <th><b>12:25PM - 1:15PM</b></th>
        <th><b>2:10PM - 3:00PM</b></th>
        <th><b>3:10PM - 4:00PM</b></th>
        <th><b>4:00PM - 4:50PM</b></th>
    </tr>
    <tr>
        <th><b>MonDay</b></th>
        <td colspan="2" style="background-color: beige;">Discrete Math
Lab</td>
        <td rowspan="5" style="background-color: rgb(238, 255,
176);">s<br>h<br>o<br>r<br>t<br>    <br>b<br>r<br>e<br>a<br>k<b></b></td>
        <td>Modren physics</td>
        <td>Object oriented programming</td>
        <td>glimpse of glorious india</td>
        <td rowspan="5" style="background-color: rgb(241, 255,
160);">L<br>u<br>n<br>c<br>h<br>    <br>b<br>r<br>e<br>a<br>k</n></td>
        <td>glimpse of glorious india</td>
        <td rowspan="5" style="background-color: rgb(192, 253,
155);">s<br>h<br>o<br>r<br>t<br>    <br>b<br>r<br>e<br>a<br>k<b></b></td>
        <td colspan="2" style="background-color: rgb(180, 205,
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        <td>Linear Algebra</td>
        <td>Discrete mathematics</td>
        <td style="color:red ;">Councelling</td>
        <td style="background-color: beige;">Manufacturing practice
Lab</td>
        <td colspan="2" style="background-color: beige;">Manufacturing
practice Lab</td>
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        <th><b>Wednesday</b></th>
        <td>Modren Physics</td>
        <td>User interface design</td>
        <td colspan="2" style="background-color: beige;">Linear
algebra Lab</td>
        <td style="color: red;">Councelling</td>
        <td>Library</td>
        <td>Glimpse of glorious india</td>
        <td>library/sports</td>
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    <tr>
        <th><b>Thursday</b></th>
        <td>Object oriented programming</td>
        <td>Discrete Mathematics</td>
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oriented programming Lab</td>
        <td>Linear algebra</td>
        <td>Library</td>
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    <th><b>Friday</b></th>
    <td>Object oriented programming</td>
    <td>Discrete mathematics</td>
    <td>Modren physics</td>
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    <td>library</td>
    <td colspan="2">Library/sports</td>
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        <th>Course code</th>
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        <th> L-T-P</th>
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        <th>Faculty Name</th>
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    <td> Dr.Srivivasa Rao Thota</td>
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        <td>23MAT117</td>
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        <td> Dr. Balaji TK</td>
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        <td>03</td>
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        <td> Mr. Barge Bharadwaj</td>
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        <td> 2-0-1</td>
        <td>02</td>
        <td>Dr. Siva Panuganti</td>
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        <td> </td>
        <td>Total (15L + 1Tut +12P =28 hrs)</td>
        <td>15-1-12</td>
        <td>21</td>
        <td> </td>
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<h1 style="text-align: center;">Syllabus</h1>
<h2 style="color: rgb(0, 0, 0);">Discrete mathematics</h2>

<h2 style="color: rgb(0, 0, 0);">Linear Algebra</h2>

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</body>
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OUTPUT :

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Syllabus

Discrete mathematics

Syllabus

Unit 1
Logic, Mathematical Reasoning and Counting: Logic, Propositional Equivalence, Predicate and Quantifiers, Theorem Proving, Recursive Definitions, Recursive Algorithms, Basics of Counting, Pigeonhole Principle, Permutation and Combinations.

Unit 2
Relations and Their Properties: Representing Relations, Closure of Relations, Partial Ordering, Equivalence Relations and partitions.
Advanced Counting Techniques and Relations: Recurrence Relations, Solving Recurrence Relations, Generating Functions, Solutions of Homogeneous Recurrence Relations, Divide and Conquer Relations, Inclusion-Exclusion.

Unit 3
Graph Theory: Graphs and Sub graphs, isomorphism, matrices associated with graphs, degrees, walks, connected graphs, shortest path algorithm. Euler and Hamilton Graphs: Euler graphs, Euler's theorem, Fleury's algorithm for Eulerian trails. Hamilton cycles, Chinese-postman problem, approximate solutions of traveling salesman problem. Closest neighbour algorithm.

Lab Practice Problems: Verifications of logical statements, truth table, tautology. Recursive algorithms. Graph problems, degree, shortest path algorithm, Euler's algorithm and closest neighbour algorithm.

Textbook(s)
 Kenneth H. Rosen, "Discrete Mathematics and its Applications", Tata McGraw- Hill Publishing Company Limited, New Delhi, Sixth Edition, 2007.
 James Strayer, "Elementary Number Theory", Waveland Press, 2002.

Linear Algebra

Syllabus

Unit 1

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Linear Algebra

Syllabus

Unit 1
 Vector Spaces: Vector spaces - Sub spaces - Linear independence - Basis - Dimension; Inner Product Spaces: Inner products - Orthogonality - Orthogonal basis - Gram Schmidt Process - Change of basis - Orthogonal complements - Projection on subspace - Least Square Principle. QR- Decomposition.

Unit 2
 Linear Transformations: Linear transformation - Relation between matrices and linear transformations - Kernel and range of a linear transformation - Change of basis - Nilpotent transformations, Symmetric and Skew Symmetric Matrices, Adjoint and Hermitian Adjoint of a Matrix, Hermitian, Unitary and Normal Transformations, Self-Adjoint and Normal Transformations.

Unit 3
 Eigen values and Eigen vectors: Eigen Values and Eigen Vectors, Diagonalization, Orthogonal Diagonalization, Quadratic Forms, Diagonalizing Quadratic Forms, Conic Sections. Similarity of linear transformations - Diagonalization and its applications - Jordan form and rational canonical form. Case Studies: Applications on least square and image transformations.

User interface design

Syllabus

Unit 1
 Introduction to Internet and Web design - Working of Web - Roles of Front-end, Back-end and Full stack development - Web Server - Internet protocols - Web Hosting - HTML - HTML Tags - Create a simple Web Page - Marking up Text - Adding Links - Adding Images - SVG - Table Markup - Embedded Media - Web Based Forms - HTML Forms - CSS for Presentation - Basic Style Sheet - A CSS Style Primer - Using Style Classes - Using Style IDs - Formatting Text - Advanced Typography with CSS3 - Working with Margins, Padding, Alignment, and Floating. Responsive web design, Introduction to version control systems.

Unit 2
 CSS Box Model and Positioning - Creating Layouts Using Modern CSS - Backgrounds and Borders - CSS Transformations and Transitions - Animating with CSS and the Canvas - Dynamic Web Pages - Web Scripting - JavaScript programming for Web Applications - Including JavaScript in HTML - HTML5 Form Controls and Validation - Document Object Model - DOM manipulation and events - Event Handlers

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Manufacturing practice

Syllabus

- 1. Additive Manufacturing Laboratory –12 hours**
Introduction to digital manufacturing. Introduction to Additive Manufacturing - types – additive manufacturing applications - Materials for 3D printing, CAD Modelling for Additive manufacturing, Slicing and STL file generation- G code generation - 3D printing of simple geometries.
- 2. Mechanical Engineering Laboratory –12 hours**
Study of tools and equipment used for basic manufacturing processes.
Manual arc welding practice for making Butt and Lap joints - Soldering Practice
Introduction to Machine Tools and Machining Processes
- 3. Automation lab –12 hours**
Design, simulate and test pneumatic and electro-pneumatic circuits. Introduction to PLC –PLC programming for automation applications.
- 4. Automobile Engineering lab –9 hours**
Overview of automobiles – components -functioning of various sub-systems; Power train, steering system, suspension system and braking system. Introduction to electric vehicles, hybrid vehicles, alternate fuels. Introduction to E Mobility.

Object oriented programming

Syllabus

Unit1
Structured to Object Oriented Approach by Examples - Object Oriented languages - Properties of Object Oriented system – UML and Object Oriented Software Development - Use case diagrams and documents as a functional model - Identifying Objects and classes - Representation of Objects and its state by Object Diagram - Simple Class using class diagram – Encapsulation - Data Hiding - Reading and Writing Objects - Class Level and Instance Level Attributes and Methods- JIVE environment for debugging.

Unit2
Aggregation and Composition using Class Diagram – Generalization using Class Diagram – Inheritance - 'Constructor and Over Riding – Visibility – Attribute – Parameter – Package - Local and Global - Polymorphism – Overloading - Abstract Classes and Interfaces.

Unit3
Exception Handling - Inner Classes - Wrapper classes – String - and String Builder classes – Number – Math – Random - Array methods - File Streams - Serialization - Generics - Collection framework - Comparator and Comparable - Vector and ArrayList - Iterator and Iterable.

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Unit3
Exception Handling - Inner Classes - Wrapper classes – String - and String Builder classes – Number – Math – Random - Array methods - File Streams - Serialization - Generics - Collection framework - Comparator and Comparable - Vector and ArrayList - Iterator and Iterable.

Modren Physics

Syllabus

Unit1
Structured to Object Oriented Approach by Examples - Object Oriented languages - Properties of Object Oriented system – UML and Object Oriented Software Development - Use case diagrams and documents as a functional model - Identifying Objects and classes - Representation of Objects and its state by Object Diagram - Simple Class using class diagram – Encapsulation - Data Hiding - Reading and Writing Objects - Class Level and Instance Level Attributes and Methods- JIVE environment for debugging.

Unit2
Aggregation and Composition using Class Diagram – Generalization using Class Diagram – Inheritance - 'Constructor and Over Riding – Visibility – Attribute – Parameter – Package - Local and Global - Polymorphism – Overloading - Abstract Classes and Interfaces.

Unit3
Exception Handling - Inner Classes - Wrapper classes – String - and String Builder classes – Number – Math – Random - Array methods - File Streams - Serialization - Generics - Collection framework - Comparator and Comparable - Vector and ArrayList - Iterator and Iterable.

Glimpses of glorious india

Syllabus

Face the Brutes, Role of Women in India, Acharya Chanakya, God and Iswara, Bhagavad Gita: From Soldier to Samsarin to Sadhaka, Lessons of Yoga from Bhagavad Gita, Indian Soft powers, Preserving Nature through Faith, Ancient Indian Cultures (Class Activity), Practical Vedanta, To the World from India, Indian Approach to Science.

Textbook(s)
"Glimpses of Glorious India", In house publication (In print).

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Explanation:

1. `<table>` = the `<table>` tag is used to arrange data into rows and columns.
2. `<tr>` = Creates a horizontal row.
3. `<th>` = Defines a header cell (usually bold and centred by default).
4. `<td>` = the `<td>` tag stands for table Data.
5. `<b>` = Used for bold text.
6. `colspan` attribute = It is used to merge multiple columns across horizontally.
7. `rowspan` attribute = It is used to merge multiple rows across vertically.

WEEK -4

1a)Creating a collage registration form using HTML.

CODE:

```
DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Week 4 - a</title>
  <style>
    table,th,td{
      border: 2px solid black;
      border-collapse: collapse;
      th,td{
        background-color: bisque;
      }
    }
  </style>
</head>
<body>
  <form>
    <h1 align="center" > Admmission Registration</h1>
    <table align="center" width="500">
      <tr>
        <td> <label for="fname">Full Name:</label></td>
        <td><input id="fname" name="fname" type="text" placeholder="Full
Name" required></td>

      </tr>
      <tr>
        <td> <label for="email">Email:</label></td>
        <td><input id="email" name="email" type="text"
placeholder="Email"></td>

      </tr>
      <tr>
        <td> <label for="phnum">Phone Number:</label></td>
        <td><input id="phnum" name="phnum" type="text" placeholder="Phone
Number"></td>

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      <tr>
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        <td> <label for="gender">Gender:</label></td>
        <td><input id="gender" name="gender" type="radio" value="Male"
>Male
        <input id="gender" name="gender" type="radio" value="Female"
>Female
        <input id="gender" name="gender" type="radio" value="Other"
>Other</td>
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    <tr>
        <td><label for="caf">Course applying for:</label></td>
        <td><select name="caf" id="caf">
            <option value="CSE">CSE</option>
            <option value="CCE">CCE</option>
            <option value="AIE">AIE</option>
        </select></td>
    </tr>
    <TR>
        <td><label for="hobby">Hobbies:</label></td>
        <td><input type="checkbox" value="Sports">Sports
        <input type="checkbox" id="hobby" name="hobby" value="Music">Music
        <input type="checkbox" id="hobby"
name="hobby" value="Reading">Reading</td>
    </TR>

    <tr>
        <td> <label for="date">Date Of Birth:</label></td>
        <td><input type="date" id="date " name="date"></td>
    </tr>
    <tr>
        <td><label for="address">Address:</label></td>
        <td><textarea name="address" id="address" width="100"
height="100"></textarea></td>

    </tr>
    <tr>
        <td colspan="2" align="center"><input type="submit">
        <input type="reset"></td>

    </tr>

</table>
</form>

</body>
</html>

```

OUTPUT:

Admission Registration	
Full Name:	Full Name
Email:	Email
Phone Number:	Phone Number
Gender:	☐ Male ☐ Female ☐ Other
Course applying for:	CSE
Hobbies:	☐ Sports ☐ Music ☐ Reading
Date Of Birth:	dd-mm-yyyy
Address:	
Submit Reset	

2b) Create two different forms in one webpage using Iframes.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Iframes</title>
</head>
<style>
  .aa{
    display: flex;
    flex-wrap: wrap;
    height: 100vh;
  }
  iframe{
    flex: 1;
  }
</style>
<body>
  <h1 align="center">Fill the forms</h1>
```

```

<div class="aa">

    <iframe src="WEEK4C.html" style="width: 500; height: 500;
border:none"></iframe>

    <iframe src="WEEK4D.html" style="width: 500; height: 500;
border:none"></iframe>
</div>
</body>
</html>

```

OUTPUT:



Fill the forms

User Login

Username:

Password:

User Registration

Full Name:	
Email:	
Phone Number:	
Password:	
Register	Reset

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>User login</title>
    <style>
        table,th,td{
            border-radius: 10px;
            border-color: rgb(10, 115, 115);
            backdrop-filter: blur(80px);
        }
    </style>
</head>
<body>
    <h1 align="center">User Login</h1>

```

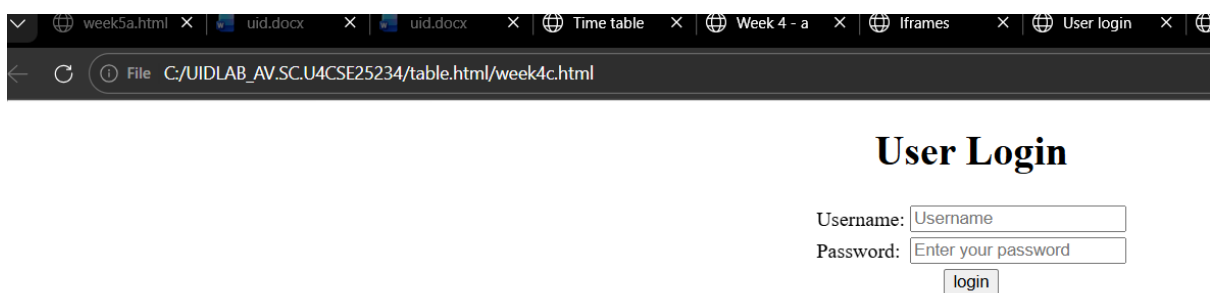
```

        <table width="200" align="center" backdrop-filter:blur>
            <form>
                <tr>
                    <td style="border-top-left-radius: 9px;"><label
for="uname">Username:</label></td>
                    <td style="border-top-right-radius: 9px;"><input type="text"
id="uname" name="uname" placeholder="Username"></td>
                </tr>
                <tr>
                    <td style="border-bottom-left-radius: 9px;"><label
for="pass">Password:</label></td>
                    <td style="border-bottom-right-radius: 9px;"><input
type="text" name="pass" id="pass" placeholder="Enter your password"></td>
                </tr>
                <tr>
                    <td colspan="2" align="center"><input type="button"
value="login" onclick="alert('login succesful')"></td>
                </tr>
            </form>
        </table>

</body>
</html>

```

OUTPUT:



The screenshot shows a web browser window with multiple tabs. The active tab displays a file from the local drive. The web page has a title "User Login" and a form with two input fields and a button. The first input field is labeled "Username:" and contains the text "Username". The second input field is labeled "Password:" and contains the text "Enter your password". Below the password field is a button labeled "login".


```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>User Registration</title>
  <style>
    table,th,td{
      border: 2px solid black;
      border-collapse: collapse;

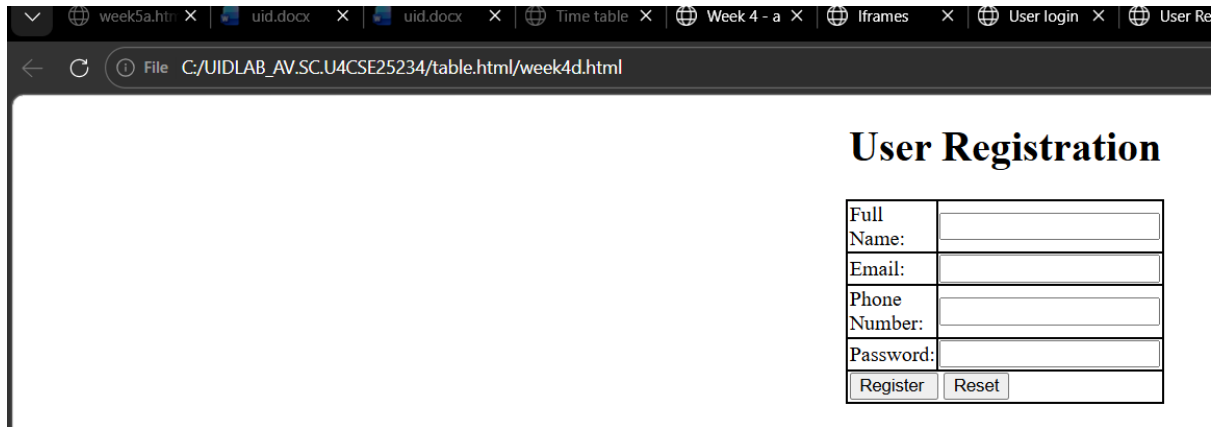
    }
  </style>
</head>
<body>

  <h1 align="center" >User Registration</h1>
  <form>
    <table align="center" width="200">
      <tr>
        <td><label for="fname">Full Name:</label></td>
        <td><input type="text" id="fname" name="fname"></td>
      </tr>
      <tr>
        <td><label for="mail">Email:</label></td>
        <td><input type="text" id="mail" name="mail"></td>
      </tr>
      <tr>
        <td><label for="phnum">Phone Number:</label></td>
        <td><input type="text" id="phnum" name="phnum"></td>
      </tr>
      <tr>
        <td><label for="pass">Password:</label></td>
        <td><input type="text" name="pass" id="pass"></td>
      </tr>
      <tr>
        <td colspan="2"><input type="button" value="Register "
onclick="alert('register successful')">
        <input type="reset" >
      </td>
      </tr>
    </table>
  </form>

```

```
</body>  
</html>
```

OUTPUT:



The screenshot shows a web browser window with multiple tabs. The active tab displays a file path: C:/UIDLAB_AV.SC.U4CSE25234/table.html/week4d.html. The page content features a heading "User Registration" followed by a form with the following fields: Full Name, Email, Phone Number, and Password. Below these fields are two buttons: "Register" and "Reset".

User Registration	
Full Name:	
Email:	
Phone Number:	
Password:	
Register	Reset

Explanation:

1. `<label>` = Provides a clickable text description that tells the user what information to enter.
2. `<input>` = Creates an interactive field where the user can actually type data or select options.
3. `<iframe>` = Acts as a window to display another website or HTML document inside your current page.
4. `<centre>` = Used to align content to the middle of the page.
5. `Src` = Tells the `<iframe>` exactly which file to display inside the window.

6. Type = Defines what kind of input to show (e.g., text for names, password to hide characters, or submit for a button).
7. Id = A unique name for an element that connects it to a <label>.
8. Name = The "key" or label sent to the server so it knows which piece of data belongs to which field.
9. For = Used in a <label> to link it to the id of an <input>, making the label clickable.

WEEK – 5

5a) Aim: creating a card layout using tag consider three cards and a button using html and css.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    body{
      background-color: #ffffff;

    }
    h1{
      text-align: center;
      font-family: 'Lucida Sans', 'Lucida Sans Regular', 'Lucida
Grande', 'Lucida Sans Unicode', Geneva, Verdana, sans-serif;
    }
    .container {
      display: flex;
      justify-content: center;
      gap: 25px;
      padding: 40px;
      background-color: #e9ecef;
      border-radius: 15px;
      margin: 20px;
    }

    .card:hover {
      background: #ffffff;
      padding: 20px;;
      max-width: 300px;
      border-radius: 10px;
      box-shadow: 0 4px 6px rgba(0,0,0,0.1);
      text-align: center;
      font-family: sans-serif;
    }
    .img{
      width: 75%;
      border-radius: 15px;
    }
  </style>
</head>
<body>
```

```

.click-me-box {
  display: block;
  margin: 15px auto 0;
  padding: 10px;
  background-color: #5148f9;
  color: white;
  border: none;
  border-radius: 55px;
  cursor: pointer;
  font-weight: bold;
}

.click-me-box:hover {
  background-color: #7094ff;
}
</style>
</head>
<body>
  <h1>IPL</h1>

  <div class="container">

    <div class="card">
      <h3>CSK</h3>
      
      <p>Chennai super kings is one of the most succesful teams in the
iplt, Known for its consistency and strong team sprit.</p>
      <button class="click-me-box">Click Me</button>
    </div>

    <div class="card">
      <h3>MI</h3>
      
      <p>Mumbai indians is most successful teams in iplt, The team is
known for smart strategy and never give up attitude.</p>
      <button class="click-me-box">Click Me</button>
    </div>

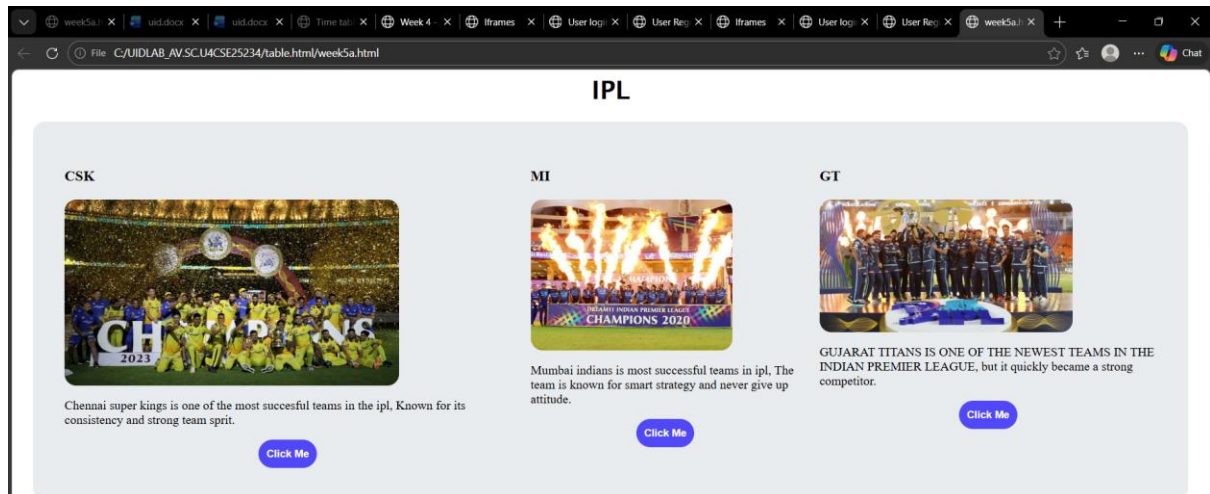
    <div class="card">
      <h3>GT</h3>
      
      <p>GUJARAT TITANS IS ONE OF THE NEWEST TEAMS IN THE INDIAN PREMIER
LEAGUE, but it quickly became a strong competitor.</p>
      <button class="click-me-box">Click Me</button>
    </div>

  </div>

```

```
</body>
</html>
```

OUTPUT:



Explanation:

1. `<div>` = A `<div>` is an invisible container used to group HTML elements together so you can style them with CSS or manipulate them with JavaScript as a single unit.
2. `<button>` = we use `<button>` element for creating button.
3. Class = A class is a reusable identifier used to apply the same CSS styles.
4. Display : flex = Display flex is a CSS property used to easily align, gap, and distribute space among items within a container.
5. Gap = Gap is the external space between multiple items inside a Flex or Grid container.

6. `Padding = Padding` is the internal space inside an element (between its border and its content).
7. `Cursor : pointer = Cursor pointer` is a CSS property used to change the mouse icon into a hand with a pointing finger.
8. `Margin = Margin` is a CSS property used to create empty space around an element.
9. `box-shadow: 0 4px 6px rgba(0,0,0,0.1) = box-shadow` is used for creating a shadow around a box.

This page was created using internal css

5b) Aim: Navigation using iframe (mini website) upon clicking each buttons the details about that button should be display on the same page.

CODE:

```
Index.html;(main)
Name = P. Srishanth , Roll no = 25234 , Section-C
<!DOCTYPE html>
<html>
  <head>
    <style>
      body{
        background-color: aliceblue;
        font-family:Arial;
      }
      .menu{
        display:flex;
        background-color: #526272;
        padding: 10px;
        text-align: center;
        justify-content: center;
        border-radius: 15px;
      }
      .menu a{
        color:white;
        margin:0 15px;
        text-decoration:none;
        background:black;
        padding:8px 12px;
        text-align: center;
        align-items: center;
        align-content: center;
        border-radius: 15px;
      }
      h1{
        text-align:center;
        background:#79ecfb;
        color:rgb(0, 0, 0);
        padding:20px;
        font-family: cursive;
        border-radius: 15px;
      }
      iframe{
        width:100%;
```



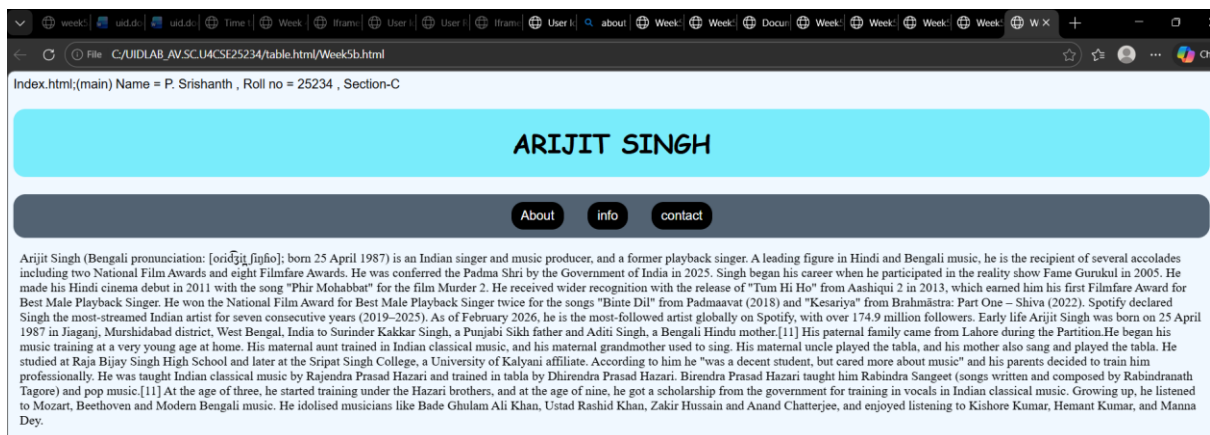
```

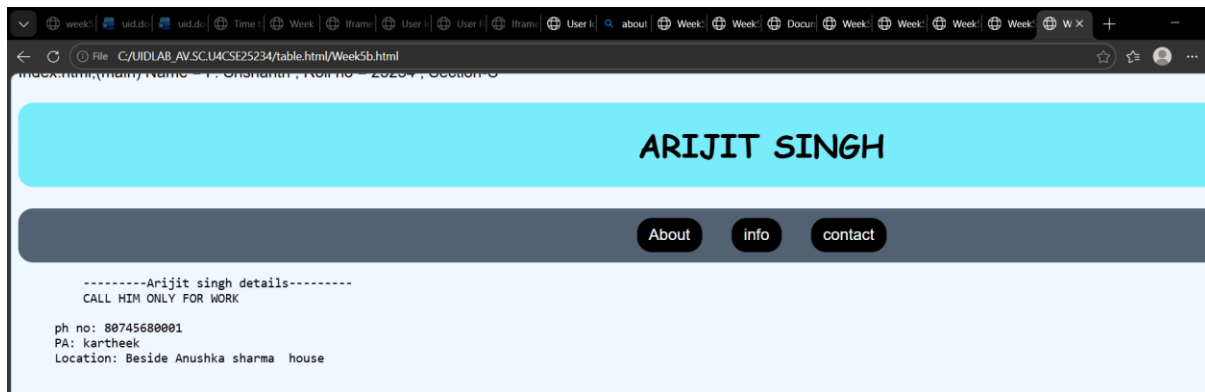
        height:4000px;
        border:none;
    }

</style>
</head>
<body>
    <h1 >ARIJIT SINGH</h1>
    <div class="menu">
        <a href="about.html" target="frame" class="menu a">About</a>
        <a href="home.html" target="frame" class="menu a">info</a>
        <a href="contact1.html" target="frame" class="menu a">contact</a>
    </div>
    <iframe name="frame"></iframe>
</body>
</html>

```

OUTPUT:





Explanation:

1. `<div>` = A `<div>` is an invisible container used to group HTML elements together so you can style them with CSS or manipulate them with JavaScript as a single unit.
2. `<button>` = we use `<button>` element for creating button.
3. Class = A class is a reusable identifier used to apply the same CSS styles.
4. Display : flex = Display flex is a CSS property used to easily align, gap, and distribute space among items within a container.
5. Gap = Gap is the external space between multiple items inside a Flex or Grid container.
6. Padding = Padding is the internal space inside an element (between its border and its content).
7. Cursor : pointer = Cursor pointer is a CSS property used to change the mouse icon into a hand with a pointing finger.
8. Margin = Margin is a CSS property used to create empty space around an element.
9. `box-shadow: 0 4px 6px rgba(0,0,0,0.1)` = box-shadow is used for creating a shadow around a box.

10.Display : inline-block = Display inline-block is a CSS property that blends the best of both worlds: it allows elements to sit side-by-side.

This code is written using internal css. And all the other three codes are combined using `<a href >` in the main code.

5C). Aim: Design amrita login page by showcampus on home page. upon clicking on each campus the details about each campus should be display on the same page use html and css to design a stylish webpage.

CODE:

```
Main.html:
Name = p.Srishanth , Roll no = 25234 , Section-C
<!DOCTYPE html>
<html>
  <head>
    <title>login webpage</title>
    <style>
      body{
        background-image: url(img.jpg);
        background-size: cover;
      }
      nav{
        background:#0b5ea6;
        text-align:center;
        padding:10px;
      }
      nav a{color:yellow;
        margin:12px;
        text-decoration:none;
        font-weight:bold;
        padding:5px;
      }
      .heading{
        font-size: 35px;
        color: white;
        background-color: rgb(195, 81, 81);
        text-align: center;
        padding:12px;
        border-radius: 15px;
      }
      .main-container {
        width: 1150px;
        height: 50px;
      }
      .login-box {
        float: left;
        width: 360px;
        background-color: #006400;
        padding: 35px;
```

```

        color: white;
        border-radius: 15px;
        opacity: 0.8;
        border: 1px solid rgba(255,255,255,0.3);
        height: 450px;
    }
    .login input[type="email"],
    .login input[type="password"]{
        width:90%;
        padding:6px;
        margin:6px 0;
    }
    .button{
        border-radius: 15px;
        background-color: #501df7;
        color: white;
        width: 75px;
    }
    iframe{
        flex:1;
        margin-left:20px;
        height:520px;
        border:none;
        background:rgb(181, 227, 255);
        border-radius:10px;
        width: 950px;
        opacity: 0.9;
    }
</style>
</head>
<body>
    <div class="heading">Amrita Vishwa Vidyapeetham</div>
    <br>

    <nav>
        <a href="amaravati.html" target="campus">AMARAVATI</a>
        <a href="amritapuri.html" target="campus">AMRITAPURI</a>
        <a href="bengaluru.html" target="campus">BENGALURU</a>
        <a href="chennai.html" target="campus">CHENNAI</a>
        <a href="coimbatore.html" target="campus">COIMBATORE</a>
        <a href="kochi.html" target="campus">KOCHI</a>
        <a href="mysuru.html" target="campus">MYSURU</a>
    </nav>
    <div class="login-box">
        <h2>Login</h2>
        Email<br>
        <input type="email"><br>
        Password <br>

```

```

        <input type="password">
        <div class="remember">
            <input type="checkbox"><label>Save me</label>
        </div>
        <input type="button" value="Login" class="button" >
        <br><br>
        <a href=" " >Forgot Password?</a><br>
        No account? <a href=" " >Register</a>
    </div>
    <iframe name="campus" src="amaravati.html"></iframe>

</body>
</html>
Amaravati.html:
<!DOCTYPE html>
<html>
    <body style=" text-align:center;">
        <h1 style="color:#a00000;">Amaravati Campus</h1>
        <p>
            One of the newer additions, this 200-acre campus in Andhra Pradesh is
            designed as a modern tech-and-business-centric educational center.
        </p>
        <p>
            nirf ranking : 8th for university<br>
            Popular Courses: CSE, AI, Data Science<br>
            Facilities: Library, Smart Classrooms, Hostel<br>
            Focus: Innovation and Research
        </p>
        
    </body>
</html>
Amritapuri.html:
<!DOCTYPE html>
<html>
    <body style="text-align:center;">
        <h1 style="color:#ff0b0b;">Amritapuri Campus</h1>
        <p>
            Located near the headquarters of the Mata Amritanandamayi Math, it's
            famous for blending high-end research with a spiritual, seaside atmosphere.
        </p>
        <p>
            nirf ranking : 8th for university<br>
            Popular Courses: Engineering, Arts, Sciences<br>
            Facilities: Research Labs, Hostel,Beautiful nature <br>
            Focus: Academic Excellence with Values
        </p>

```

```

        
</body>
</html>
Bengaluru.html
<!DOCTYPE html>
<html>
    <body style="text-align:center;">
        <h1 style="color:#a00000;">Bengaluru Campus</h1>
        <p>
            Located in India's tech hub, Bengaluru campus offers advanced
            engineering programs with strong industry connections.
        </p>
        <p>
            nirf ranking : 8th for university <br>
            Popular Courses: Robotics, CSE, ECE<br>
            Facilities: Coding Labs, Innovation Centres<br>
            Focus: Industry Collaboration
        </p>
        
    </body>
</html>
Chennai.html:
<!DOCTYPE html>
<html>
    <body style="text-align:center;">
        <h1 style="color:#a00000;">Chennai Campus</h1>
        <p>
            Chennai campus provides top-quality computing and management education
            with modern infrastructure.
        </p>
        <p>
            nirf ranking : 8th for university<br>
            Popular Courses: IT, MBA, Electronics<br>
            Facilities: Digital Library, Placement Cell<br>
            Focus: Career Development
        </p>
        
    </body>
</html>
Coimbatore.html:
<!DOCTYPE html>
<html>
    <body style="text-align:center;">
        <h1 style="color:#a00000;">Coimbatore Campus</h1>
        <p>

```

```

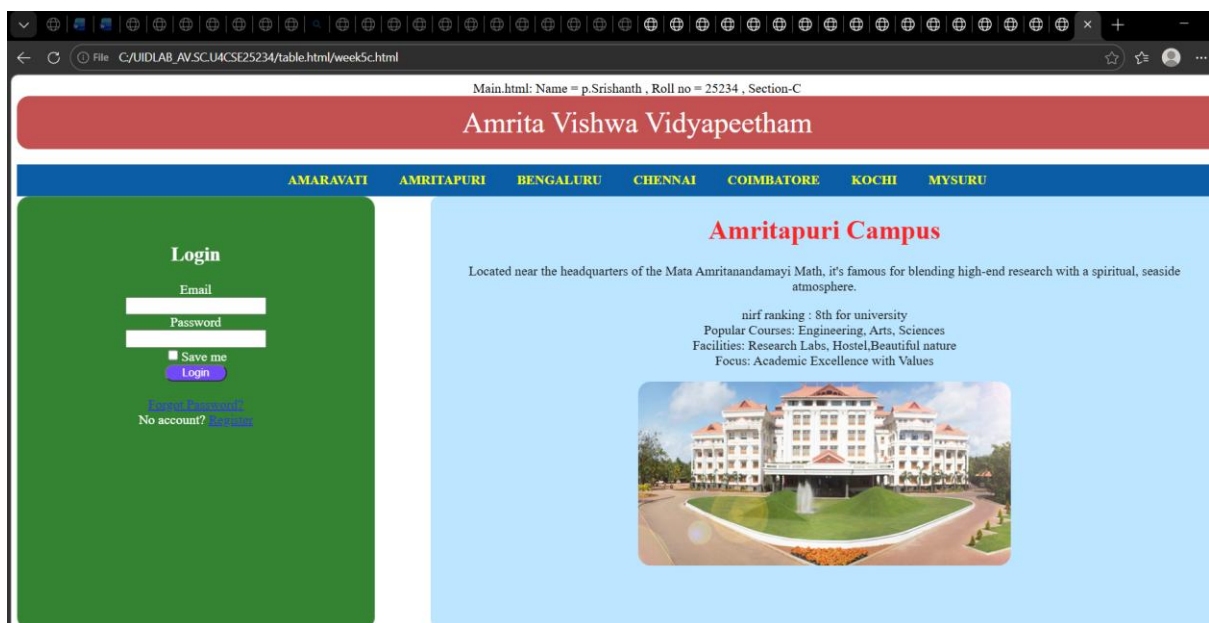
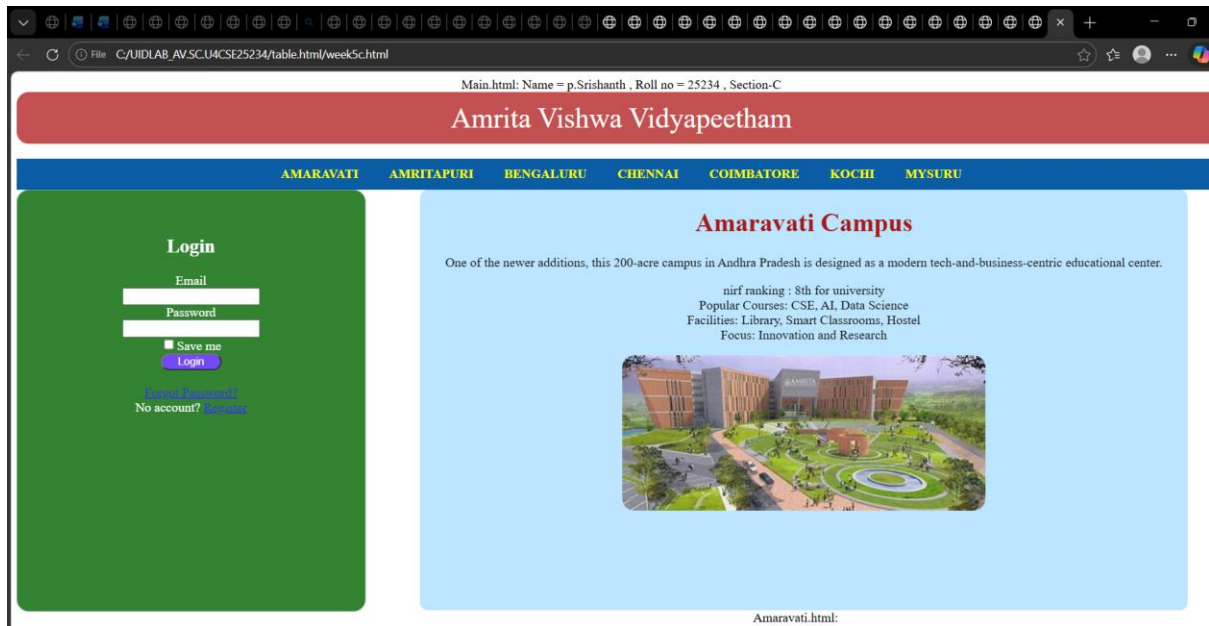
        The massive 450-acre headquarters known for its top-tier engineering
        programs and lush, eco-friendly "oasis" setting.
    </p>
    <p>
        nirf ranking : 8th for university<br>
        Popular Courses: Mechanical, Civil, CSE<br>
        Facilities: Workshops, Labs, Hostel<br>
        Focus: Core Engineering Skills
    </p>
    
</body>
</html>
Kochi.html:
<!DOCTYPE html>
<html>
    <body style="text-align:center;">
        <h1 style="color:#a00000;">Kochi Campus</h1>
        <p>
            A premier healthcare hub that houses the renowned Amrita Institute of
            Medical Sciences (AIMS)
            and a major multi-specialty hospital.
        </p>
        <p>
            nirf ranking : 8th for university<br>
            Popular Courses: MBA, Commerce, Finance<br>
            Facilities: Seminar Halls, Library<br>
            Focus: Industry Training
        </p>
        
    </body>
</html>
Mysuru.html:
<!DOCTYPE html>
<html>
    <body style="text-align:center;">
        <h1 style="color:#a00000;">Mysuru Campus</h1>
        <p>
            Mysuru campus is known for its peaceful environment,
            green campus, and quality education.
        </p>
        <p>
            nirf ranking : 8th for university<br>
            Popular Courses: Arts, Sciences, Engineering<br>
            Facilities: Hostel, Library, Sports Complex<br>
            Focus: Calm Learning Atmosphere
        </p>
        

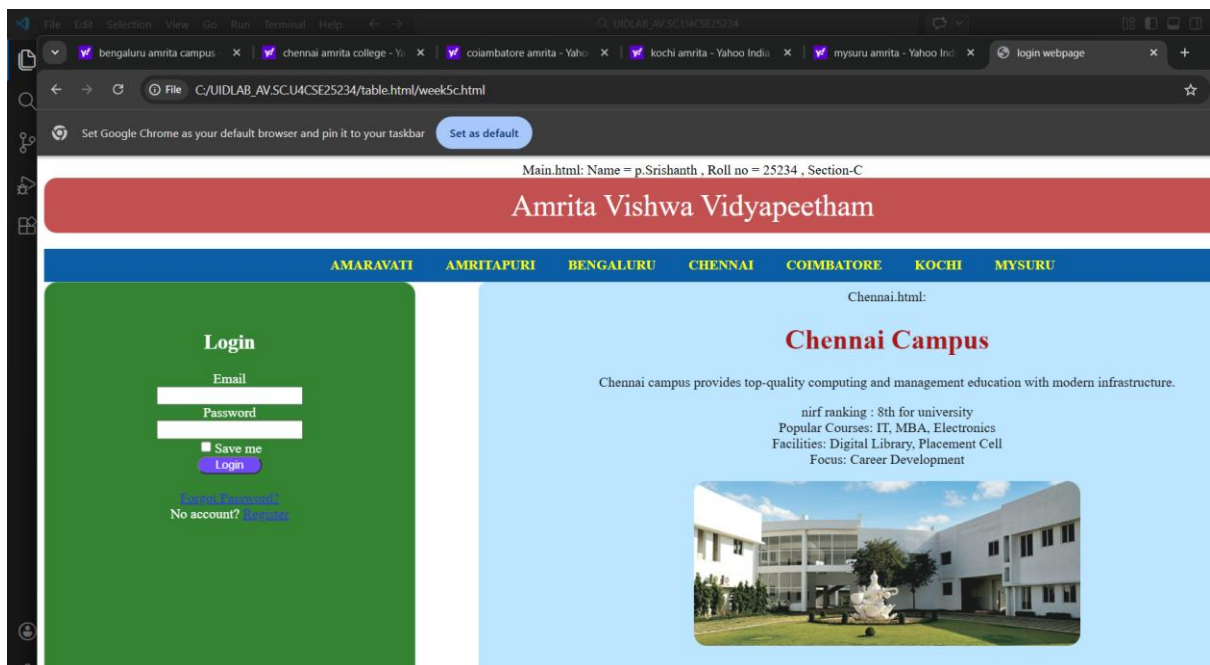
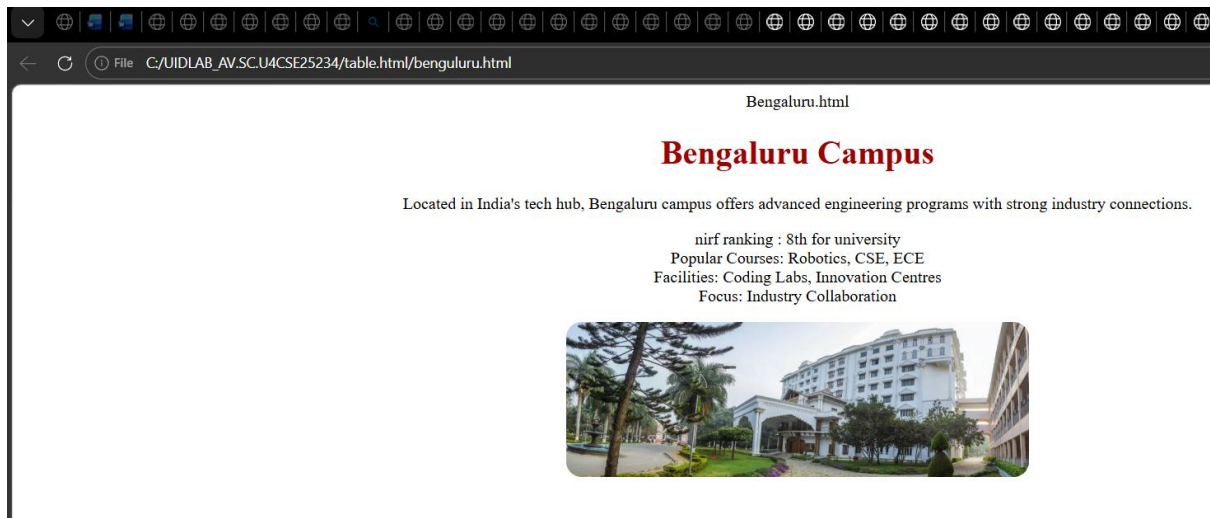
```



```
</body>
</html>
```

OUTPUT:





File C:/UIDLAB_AV.SC.U4CSE25234/table.html/coiambatore.html

Coimbatore Campus

The massive 450-acre headquarters known for its top-tier engineering programs and lush, eco-friendly "oasis" setting.

nirf ranking : 8th for university
 Popular Courses: Mechanical, Civil, CSE
 Facilities: Workshops, Labs, Hostel
 Focus: Core Engineering Skills



File C:/UIDLAB_AV.SC.U4CSE25234/table.html/week5c.html

Main.html: Name = p.Srishanth , Roll no = 25234 , Section-C

Amrita Vishwa Vidyapeetham

AMARAVATI AMRITAPURI BENGALURU CHENNAI COIMBATORE KOCHI MYSURU

Login

Email

Password

☐ Save me

Login

[Forgot Password?](#)

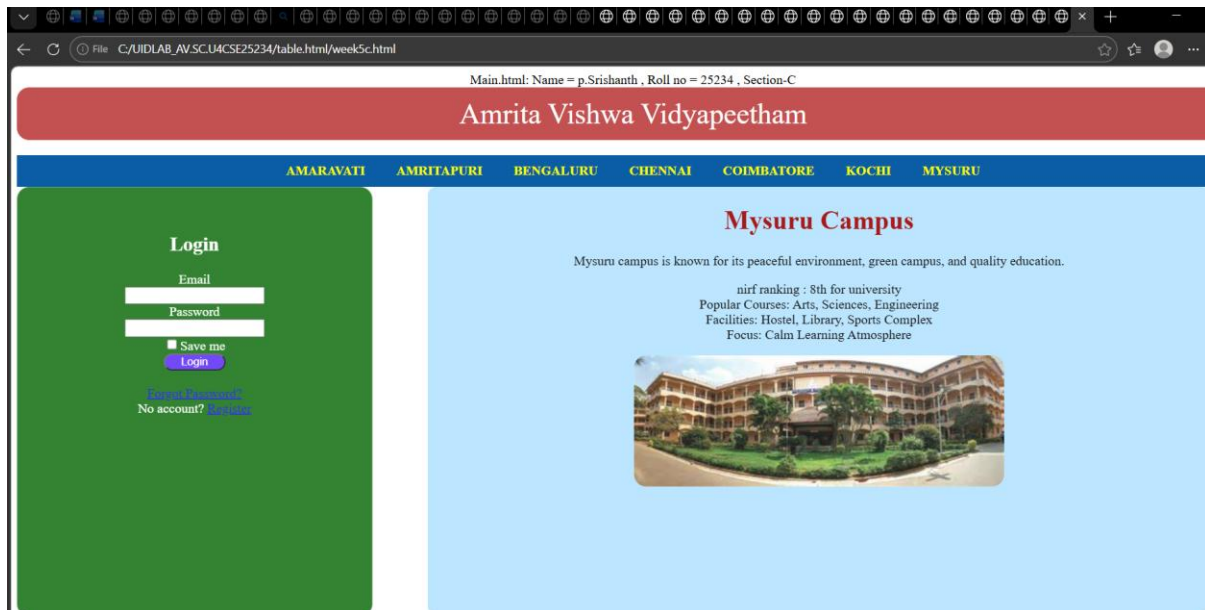
[No account? Register](#)

Kochi Campus

A premier healthcare hub that houses the renowned Amrita Institute of Medical Sciences (AIMS) and a major multi-specialty hospital.

nirf ranking : 8th for university
 Popular Courses: MBA, Commerce, Finance
 Facilities: Seminar Halls, Library
 Focus: Industry Training





Explanation:

1. `<div>` = A `<div>` is an invisible container used to group HTML elements together so you can style them with CSS or manipulate them with JavaScript as a single unit.
2. `<button>` = we use `<button>` element for creating button.
3. Class = A class is a reusable identifier used to apply the same CSS styles.
4. Display : flex = Display flex is a CSS property used to easily align, gap, and distribute space among items within a container.
5. Gap = Gap is the external space between multiple items inside a Flex or Grid container.
6. Padding = Padding is the internal space inside an element (between its border and its content).
7. Cursor : pointer = Cursor pointer is a CSS property used to change the mouse icon into a hand with a pointing finger.

8. Margin = Margin is a CSS property used to create empty space around an element.
9. box-shadow: 0 4px 6px rgba(0,0,0,0.1) = box-shadow is used for creating a shadow around a box.
10. Display : inline-block = Display inline-block is a CSS property that blends the best of both worlds: it allows elements to sit side-by-side.

This code is written using internal css. And all the other three codes are combined using `<a href >` in the main code.

Week-6

1A) Aim: Design a responsive e-commerce application using the grid layout. This application should contain different categories of products such as electronics, clothing, home-appliances.

Program:

MAIN.HTML

```
!<DOCTYPE html>
<html>
<head>
<title>My Ecommerce Store</title>

<style>

body{
font-family: Arial;
margin:0;
background:#f5f5f7;
}

.header{
background:#0f172a;
color:white;
text-align:center;
padding:20px;
font-size:28px;
}

.sale{
background:#2563eb;
color:white;
text-align:center;
padding:10px;
font-size:18px;
font-weight:bold;
}

.container{
display:flex;
min-height:80vh;
}

.sidebar{
```

```

width:200px;
background:#1e293b;
color:white;
padding:20px;
}

.sidebar h3{
margin-top:0;
}

.sidebar a{
display:block;
color:white;
text-decoration:none;
padding:10px 0;
border-bottom:1px solid #334155;
}

.sidebar a:hover{
color:#60a5fa;
}

.frame{
flex:1;
border:none;
}

.footer{
background:#0f172a;
color:white;
text-align:center;
padding:15px;
}

</style>
</head>

<body>

<div class="header">
My Ecommerce Store
</div>

<div class="sale">
Big Sale! Up to 50% Off
</div>

<div class="container">

```

```

<div class="sidebar">
<h3>Categories</h3>

<a href="week 6 elec.html" target="frame">Electronics</a>
<a href="6 cloth.html" target="frame">Clothes</a>
<a href="access.html" target="frame">Accessories</a>

</div>

<iframe class="frame" name="frame" src="week 6 elec.html"></iframe>

</div>

<div class="footer">
© 2026 My Ecommerce Store
</div>

</body>
</html>

```

ELECTRONICS.HTML

```

<!DOCTYPE html>
<html>
<head>
  <title>Electronics</title>
  <style>
    body {
      margin: 0;
      background-color: aliceblue;
      font-family: sans-serif;
    }

    .products {
      grid-area: products;
      display: grid;
      grid-template-columns: repeat(2, 1fr);
      gap: 10px;
      padding: 20px;
    }

    .product-card {
      background: white;

```



```

        border: 1px solid #ddd;
        padding: 20px;
        text-align: center;
        border-radius: 8px;
        height: auto;
        width: 600px;
    }
    .p{
        text-align: center;
        background-color: tomato;
        align-items: center;
    }
    img{
        width: 120px;
        height: auto;
        border-radius: 8px;
    }

    .button {
        background-color: blue;
        color: white;
        border: none;
        padding: 10px 20px;
        cursor: pointer;
        border-radius: 50px;
        margin-top: 10px;
    }

    .button:hover {
        background-color: #7094ff;
    }
</style>
</head>
<body>
    <div class="products">
        <div class="product-card">
            
            <h3>Iphone 17 pro max</h3>
            <p>$899</p>
            <p>12GB RAM | 1TB Storage</p>
            <button class="button">Add to Cart</button>
        </div>
        <div class="product-card">
            
            <h3>pixel9a</h3>
            <p>$1199</p>
            <p>12GB RAM | 512GB Storage</p>

```

```

        <button class="button">Add to Cart</button>
    </div>
    <div class="product-card">
        
        <h3>One Plus Nord 5</h3>
        <p>$299</p>
        <p>12GB RAM | 256GB Storage</p>
        <button class="button">Add to Cart</button>
    </div>
    <div class="product-card">
        
        <h3>Nothing 3a</h3>
        <p>$249</p>
        <p>12GB RAM | 256GB Storage</p>
        <button class="button">Add to Cart</button>
    </div>
</div>
</body>

```

CLOTHES.HTML

```

<!DOCTYPE html>
<html>
<head>
    <title>Electronics</title>
    <style>
        body {
            margin: 0;
            background-color: aliceblue;
            font-family: sans-serif;
        }

        .products {
            grid-area: products;
            display: grid;
            grid-template-columns: repeat(2, 1fr);
            gap: 20px;
            padding: 20px;
        }

        .product-card {
            background: white;

```

```

        border: 1px solid #ddd;
        padding: 20px;
        text-align: center;
        border-radius: 8px;
    }
    img{
        width: 120px;
        height: auto;
        border-radius: 8px;
    }
    .p{
        text-align: center;
        background-color: tomato;
        align-items: center;
    }

    .button {
        background-color: blue;
        color: white;
        border: none;
        padding: 10px 20px;
        cursor: pointer;
        border-radius: 50px;
        margin-top: 10px;
    }

    .button:hover {
        background-color: #7094ff;
    }
</style>
</head>
<body>
    <div class="products">
        <div class="product-card">
            
            <h3>H&M csk jersey</h3>
            <p>$39</p>
            <p>Size: S , M , XL , XXL</p>
            <button class="button">Add to Cart</button>
        </div>
        <div class="product-card">
            
            <h3>H&M Blue jean</h3>
            <p>$59</p>
            <p>Size: S , M , L , XL</p>
            <button class="button">Add to Cart</button>
        </div>
    </div>

```

```

        <div class="product-card">
            
            <h3>H&M black coat</h3>
            <p>$49</p>
            <p>Size: 28 , 30 , 32 , 34</p>
            <button class="button">Add to Cart</button>
        </div>
        <div class="product-card">
            
            <h3>USPA black pant</h3>
            <p>$52</p>
            <p>Size: 28 , 30 , 32 , 34</p>
            <button class="button">Add to Cart</button>
        </div>
    </div>
</body>

```

ACCESSORIES:

```

<!DOCTYPE html>
<html>
<head>
    <title>Electronics</title>
    <style>
        body {
            margin: 0;
            background-color: aliceblue;
            font-family: sans-serif;
        }

        .products {
            grid-area: products;
            display: grid;
            grid-template-columns: repeat(2, 1fr);
            gap: 20px;
            padding: 20px;
        }

        .product-card {
            background: white;
            border: 1px solid #ddd;
            padding: 20px;
        }
    </style>

```

```

        text-align: center;
        border-radius: 8px;
    }
    .p{
        text-align: center;
        background-color: tomato;
        align-items: center;
    }
    img{
        width: 120px;
        height: auto;
        border-radius: 8px;
    }

    .button {
        background-color: blue;
        color: white;
        border: none;
        padding: 10px 20px;
        cursor: pointer;
        border-radius: 50px;
        margin-top: 10px;
    }

    .button:hover {
        background-color: #7094ff;
    }
</style>
</head>
<body>
    <div class="products">
        <div class="product-card">
            
            <h3>Samsung Galaxy Watch 6</h3>
            <p>$169</p>
            <button class="button">Add to Cart</button>
        </div>
        <div class="product-card">
            
            <h3>Samsung Galaxy Buds 4</h3>
            <p>$249</p>
            <button class="button">Add to Cart</button>
        </div>
        <div class="product-card">
            
            <h3>Samsung Galaxy ring </h3>
            <p>$459</p>

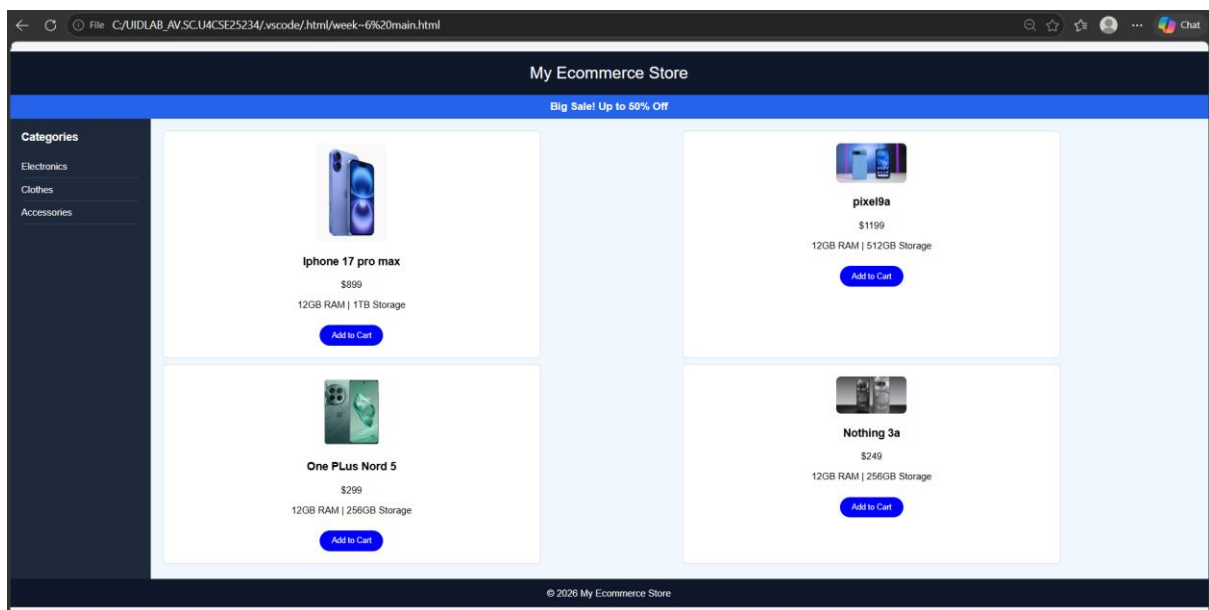
```

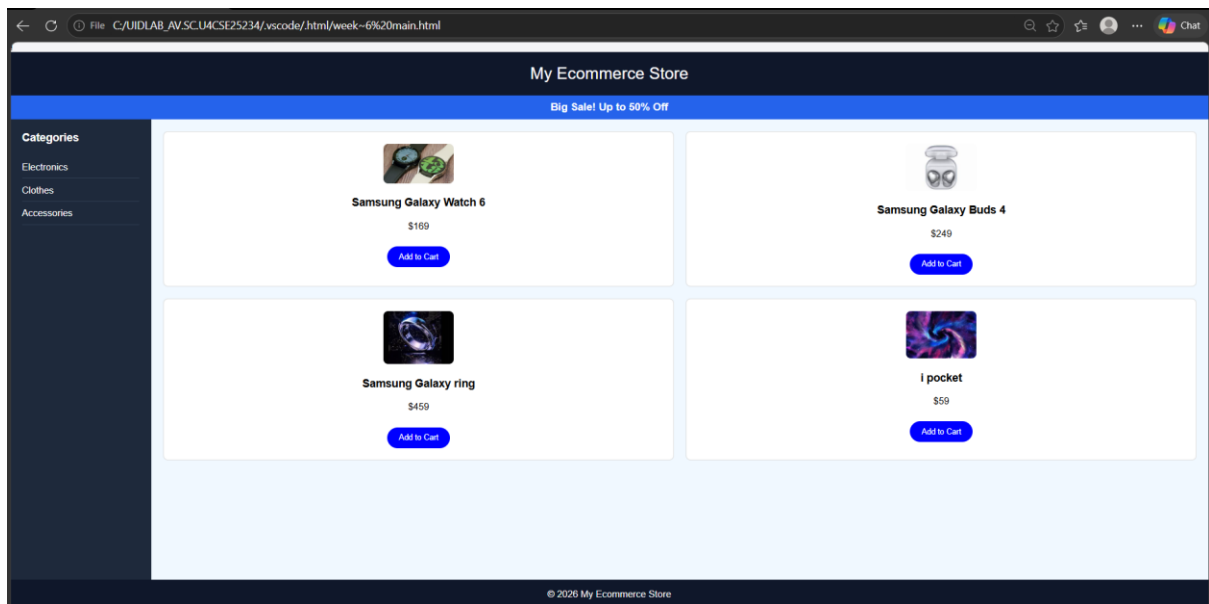
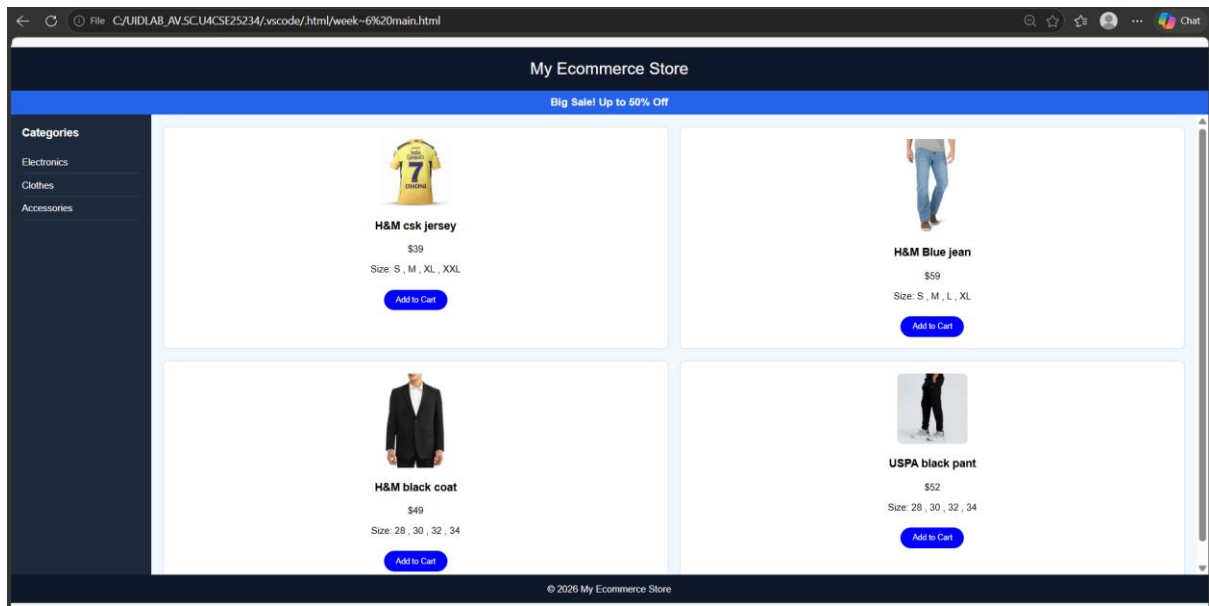
```

        <button class="button">Add to Cart</button>
    </div>
    <div class="product-card">
        
        <h3>i pocket </h3>
        <p>$59</p>
        <button class="button">Add to Cart</button>
    </div>
</div>
</body>

```

OUTPUT:





Explanation:

1. <div> = A <div> is an invisible container used to group HTML elements together so you can style them with CSS
2. <button> = we use <button> element for creating button.
3. Class = A class is a reusable identifier used to apply the same CSS styles.
4. Display : grid = Display grid is a CSS property used to turn an element into a grid container.
5. grid-template-columns: Defines how many columns you want.
6. grid-template-rows: Defines the height of your rows.
7. gap: Controls the spacing between your grid items without using margins.
8. grid-area: Assigns a name to an item so you can place it exactly where you want in the layout.
9. Padding = Padding is the internal space inside an element (between its border and its content).
- 10.<I frame>=we <iframe> tag to create link other webpages into it.
11. : we use tag for displaying the image.
12. Cursor:pointer = Cursor pointer is a CSS property used to change the mouse icon into a hand with a pointing finger.

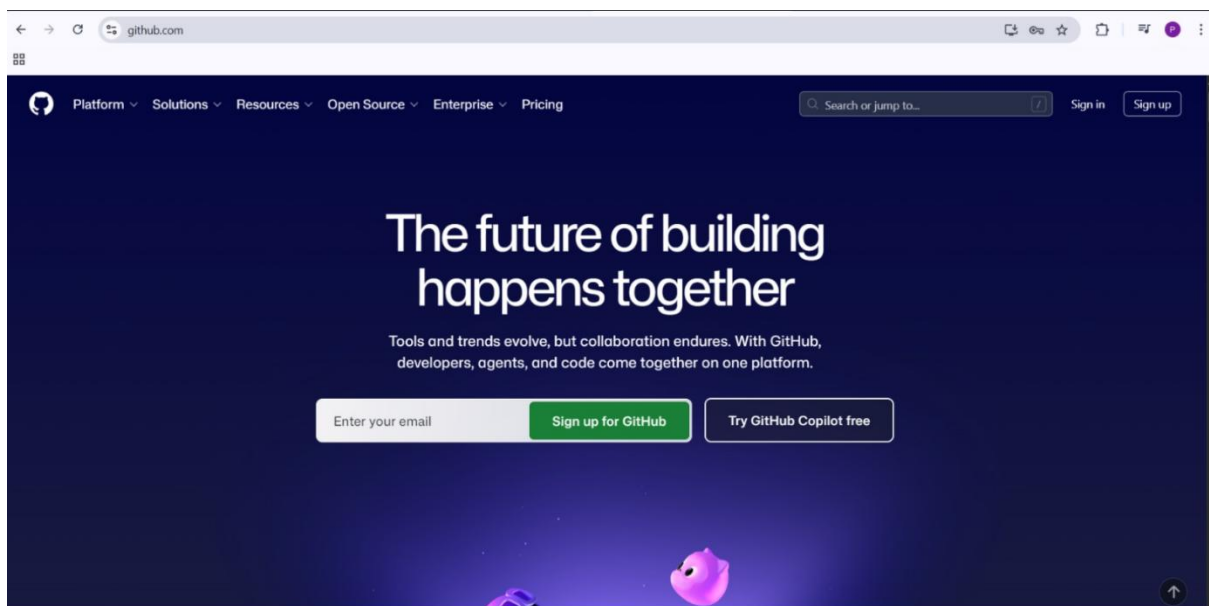
This code is written using internal css. And all the other three codes are combined using <a href > in the main code.

Week-7

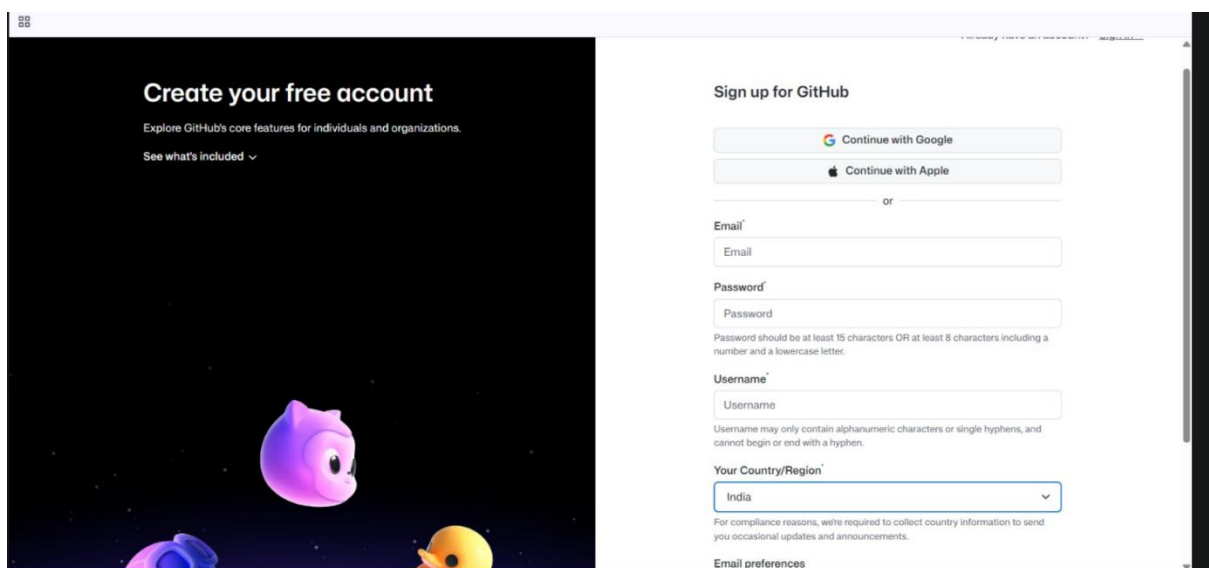
7A) Aim: create a GitHub account and create a repository in it and push all the week 6 code files in it.

Procedure :

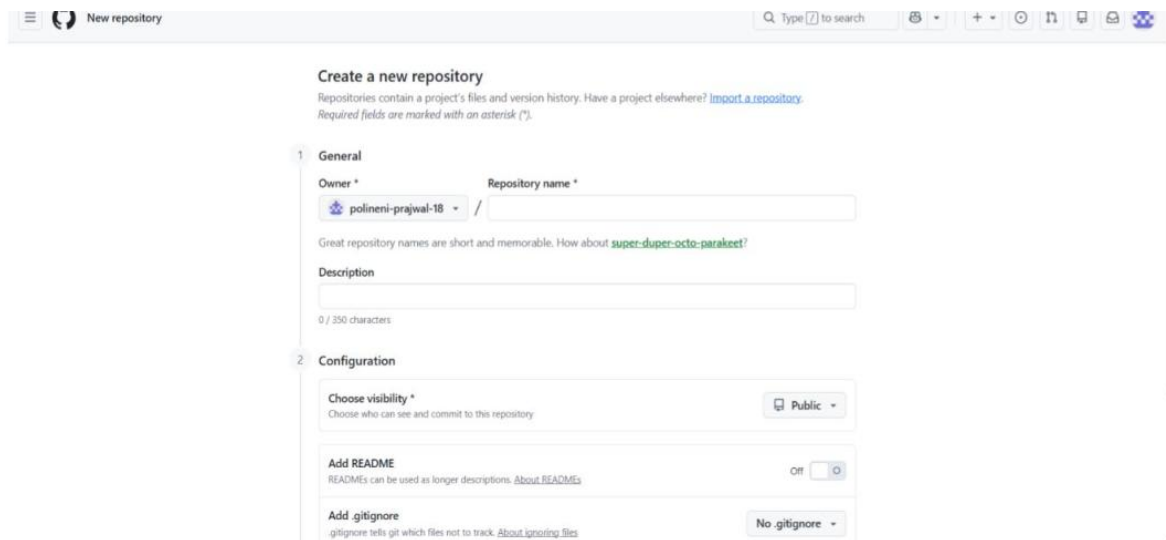
Step 1: browse for GitHub and enter it.



Step 2: login to Github and enter it.



Step3: Create a repository



The screenshot shows the 'Create a new repository' page on GitHub. The page has a header with the GitHub logo and a search bar. Below the header, there's a section titled 'Create a new repository' with a subtitle 'Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository.](#)' and a note 'Required fields are marked with an asterisk (*)'.

The form is divided into two sections: '1 General' and '2 Configuration'.

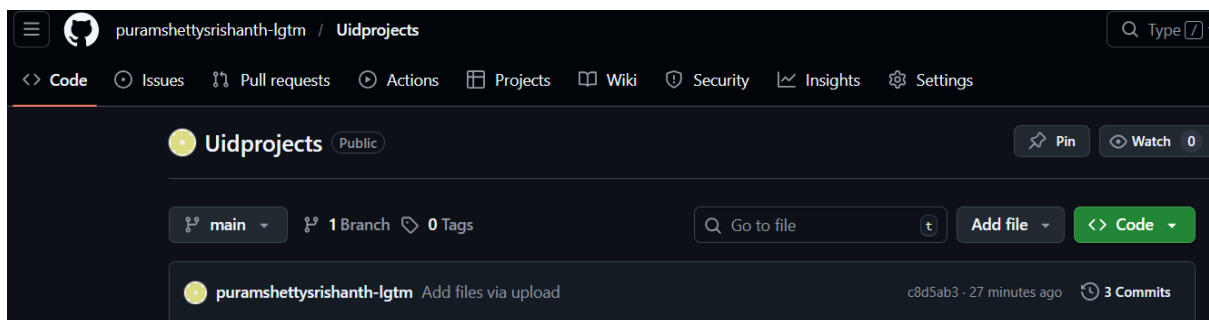
1 General

- Owner ***: A dropdown menu showing 'polineni-prajwal-18'.
- Repository name ***: An empty text input field.
- Description**: A text input field with a placeholder 'Great repository names are short and memorable. How about `super-duper-octo-parakeet`?'. Below the field, it says '0 / 350 characters'.

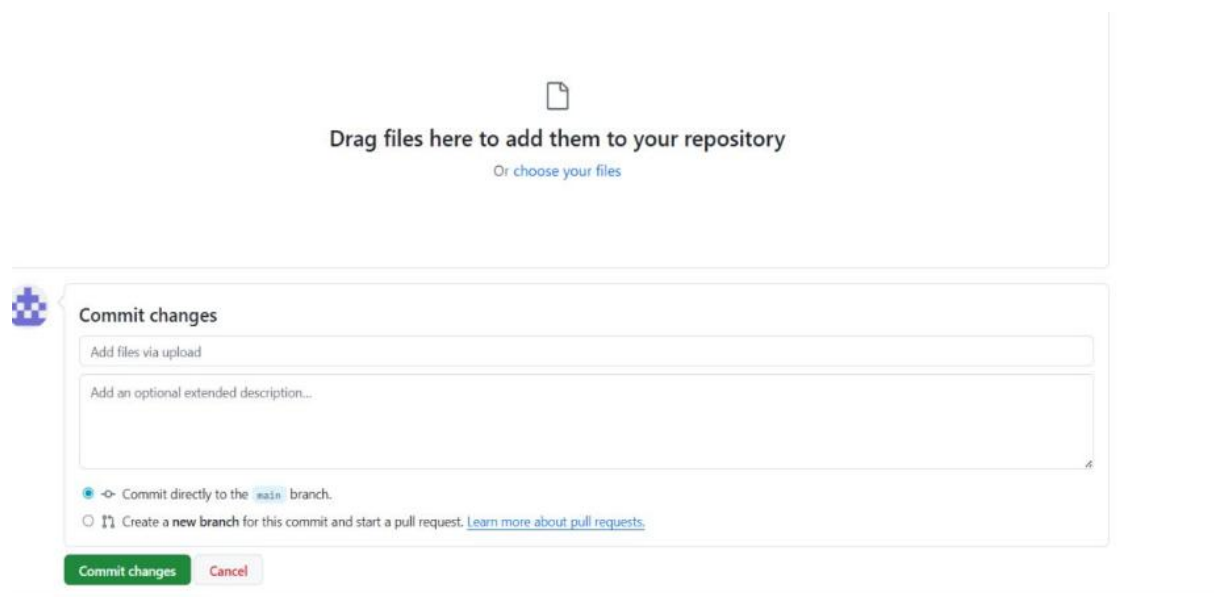
2 Configuration

- Choose visibility ***: A dropdown menu showing 'Public'.
- Add README**: A section with a subtitle 'READMEs can be used as longer descriptions. [About READMEs](#)'. It has a toggle switch set to 'Off'.
- Add .gitignore**: A section with a subtitle '.gitignore tells git which files not to track. [About ignoring files](#)'. It has a dropdown menu showing 'No .gitignore'.

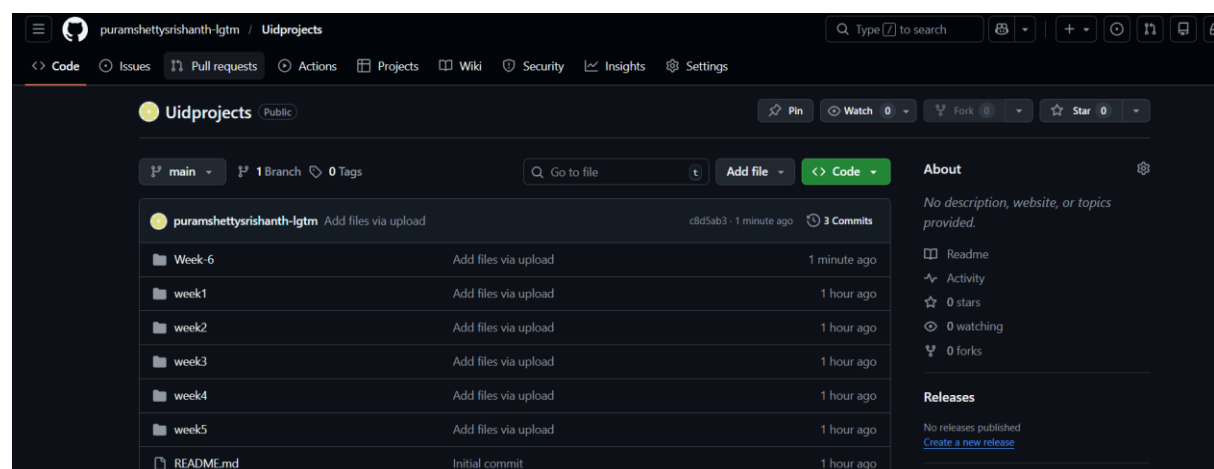
Step 4: Click on upload files and upload the files for the device storage.



Step 5: Drag files into the given box and click on the commit changes.



Step 6:



Link for repository:

<https://github.com/puramshettysrishanth-lgtm/Uidprojects.git>

WEEK-8

8A) Aim: To create a webpage that displays a square that continuously rotates 360° using css animations.

Program:

```
<!DOCTYPE html>
<html>
<head>
  <title>Rotating Square</title>
  <style>
    body {
      margin: 0;
      height: 100vh;
      display: flex;

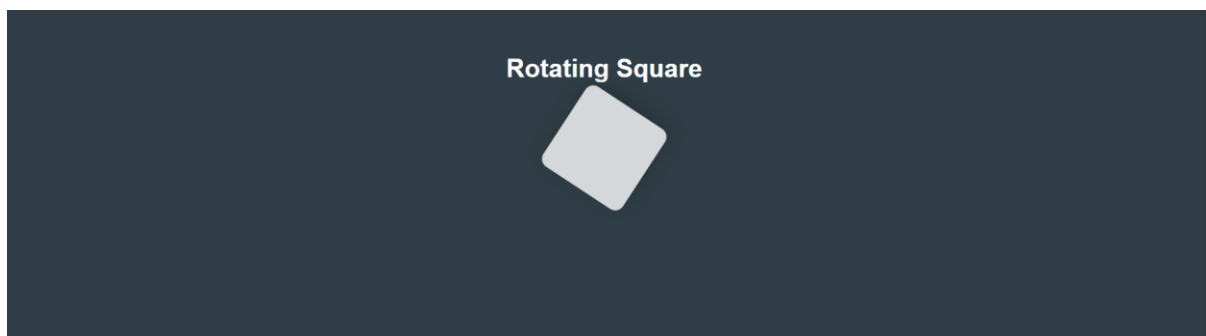
      flex-direction: column;
      justify-content: flex-start;
      align-items: center;
      background-color: #2f3e46;
      font-family: Arial, sans-serif;
      padding-top: 50px;
    }
    .square {
      width: 120px;
      height: 120px;
      background: #ffffffcc;
      border-radius: 12px;
      box-shadow: 0 8px 25px rgba(0, 0, 0, 0.2);
      animation: rotate 3s linear infinite;
    }
    @keyframes rotate {
      from {
        transform: rotate(0deg);
      }
      to {
        transform: rotate(360deg);
      }
    }
  </style>
</head>
<body>
  <div class="square"></div>
</body>
</html>
```

```

    }
  }
</style>
</head>
<body>
  <h1 style="color: #fff;">Rotating Square</h1>
  <div class="square"></div>
</body>
</html>

```

OUTPUT:



EXPLANATION:

1. `display: flex` = It is used to easily align the items and distribute space within the container.
2. `flex-direction: column` = is used to change the orientation of your layout from a horizontal row to a vertical column.
3. `Align-items: centre` = It is used to horizontally center the content because of the flex direction.
4. `<div>` = A `<div>` is an invisible container used to group HTML elements together so you can style them with CSS.
5. `@keyframes` = a keyframe tells the computer exactly how an element should look at a specific point in time.

6. Animation = Animation is shorthand property for name , duration , timing function , delay , iteration count , direction , fill mode , play state.

8B). AIM:

TO create a ball that moves up and down continuously using css animations.

Program:

```
<!DOCTYPE html>
<html>
<head>
  <title>Bouncing Ball</title>
  <style>
    body {
      margin: 0;
      height: 100vh;
      display: flex;
      flex-direction: column;
      align-items: center;
      justify-content: flex-start;
      background: #bc6208;
    }
    h1 {
      margin-top: 20px;
      color: #ffffff;
    }
    .ball {
      width: 80px;

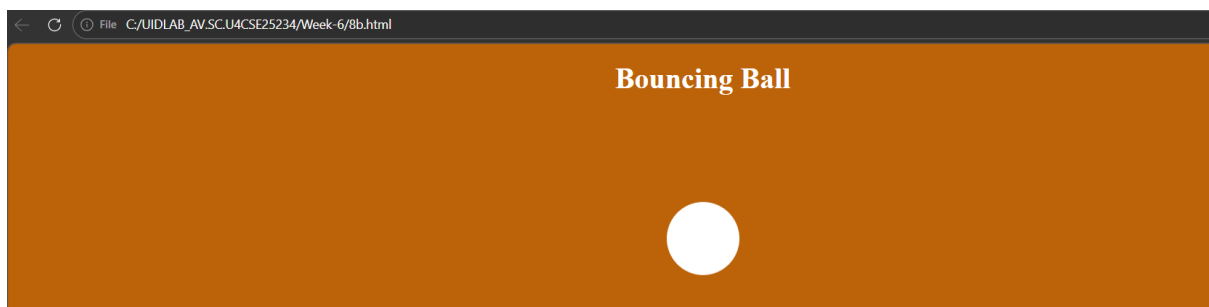
      height: 80px;
      background: #ffffff;
      border-radius: 50%;
      margin-top: 120px;
      animation: bounce 1.5s ease-in-out infinite;
    }
    @keyframes bounce {
      0% {
        transform: translateY(0);
```

```

    }
    50% {
        transform: translateY(-120px);
    }
    100% {
        transform: translateY(0);
    }
}
</style>
</head>
<body>
    <h1>Bouncing Ball</h1>
    <div class="ball"></div>
</body>
</html>

```

OUTPUT:



Explanation:

1. `flex-direction: column` = is used to change the orientation of your layout from a horizontal row to a vertical column.
2. `display: flex` = It is used to easily align the items and distribute space within the container.
3. `Align-items: centre` = It is used to horizontally center the content because of the flex direction.
4. `<div>` = A `<div>` is an invisible container used to group HTML elements together so you can style them with CSS.
5. `@keyframes` = a keyframe tells the computer exactly how an element should look at a specific point in time.

6. Animation = Animation is shorthand property for name , duration , timing function , delay , iteration count , direction , fill mode , play state.
7. We denote percentages to tell at a particular time which work should be happen.
8. The transform: translateY(550px); property is a CSS tool used to move an element vertically along the Y-axis.

8C). AIM: To create a ball that moves horizontally and rotates at same time.

Program:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Moving Rotating Circle</title>
  <style>
    body {
      margin: 0;
      height: 100vh;
      display: flex;
      flex-direction: column;
      align-items: center;
      justify-content: flex-start;
      background: #2f3e46;
      overflow: hidden;
    }
    h1 {
      margin-top: 20px;
      color: #ffffff;
    }
    .circle {
      width: 80px;
      height: 80px;
      background: #ffffff;
      border-radius: 50%;
      margin-top: 120px;
      animation: moveRotate 3s linear infinite;
    }
  </style>
</head>
<body>
  <h1>Moving Rotating Circle</h1>
  <div class="circle"></div>
</body>
</html>
```



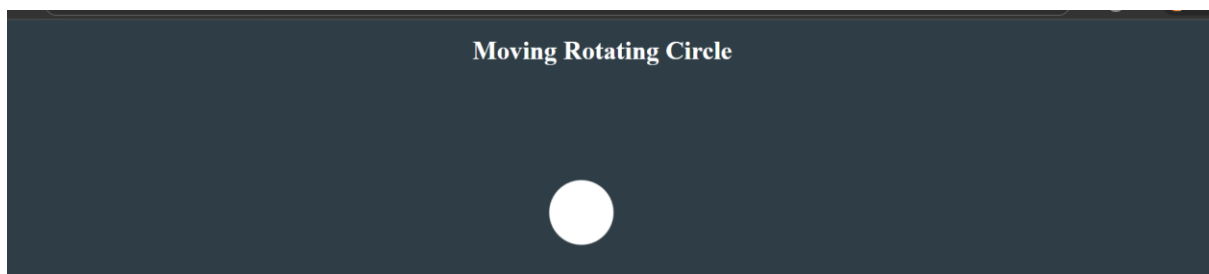
```

@keyframes moveRotate {
  0% {
    transform: translateX(-150px) rotate(0deg);
  }
  50% {
    transform: translateX(150px) rotate(180deg);
  }
  100% {
    transform: translateX(-150px) rotate(360deg);
  }
}
</style>
</head>
<body>
  <h1>Moving Rotating Circle</h1>

  <div class="circle"></div>
</body>
</html>

```

OUTPUT:



Explanation:

1. flex-direction: column = is used to change the orientation of your layout from a horizontal row to a vertical column.
2. display: flex = It is used to easily align the items and distribute space within the container.
3. Align-items: centre = It is used to horizontally center the content because of the flex direction.

4. `<div>` = A `<div>` is an invisible container used to group HTML elements together so you can style them with CSS.
5. `@keyframes` = a keyframe tells the computer exactly how an element should look at a specific point in time.
6. `Animation` = Animation is shorthand property for name , duration , timing function , delay , iteration count , direction , fill mode , play state.
7. We denote percentages to tell at a particular time which work should be happen.
8. The `transform: translateX(____px);` property is a CSS tool used to move an element vertically along the X-axis.

WEEK-9

9A) Aim: Week 9: Create a webpage using html, css, js based on the given conditions, create a button for each program, based on the button, clicked the corresponding program has to be executed and should display the logics for each button includes finding an even or odd number, prime or not, factorial of a number, Armstrong or not use appropriate design to display the output.

Program:

```
<!DOCTYPE html>
<html>
<head>
  <title>Logic Programs</title>
  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background: #1a1a2e;
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
    }
    .container {
      background: white;
      border-radius: 12px;
      box-shadow: 0 4px 20px rgba(0,0,0,0.3);
      padding: 40px;
      width: 380px;
      text-align: center;
    }
    h2 {
      margin-bottom: 20px;
      color: #333;
    }
    input {
      width: 100%;
      padding: 10px;
      font-size: 16px;
      border: 2px solid #ccc;
      border-radius: 8px;
      margin-bottom: 20px;
      box-sizing: border-box;
      text-align: center;
    }
```

```

}
input:focus {
    outline: none;
    border-color: #cacfd4;
}
.buttons {
    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 12px;
    margin-bottom: 24px;
}
button {
    padding: 12px;
    font-size: 15px;
    border: none;
    border-radius: 8px;
    cursor: pointer;
    color: rgb(250, 26, 26);
    font-weight: bold;
    background: #93de08;
    transition: opacity 0.2s;
}
button:hover { opacity: 0.85; }
.output {
    background: #c2d4e6;

    padding: 20px;
    border-radius: 8px;
    min-height: 60px;
    color: #0d6084;
    font-size: 16px;
}
</style>
</head>
<body>

<div class="container">
    <h2>Logic Programs</h2>
    <input type="number" id="numInput" placeholder="Enter a number" />
    <div class="buttons">
        <button onclick="checkEvenOdd()">Even / Odd</button>
        <button onclick="checkPrime()">Prime / Not</button>

        <button onclick="calcFactorial()">Factorial</button>
    
```

```

        <button onclick="checkArmstrong()">Armstrong</button>
    </div>
    <div class="output" id="output">Result will appear here</div>
</div>
<script>
    function getInput() {
        const val = document.getElementById('numInput').value.trim();
        if (val === '' || isNaN(val)) {
            document.getElementById('output').textContent = 'Please enter
a
valid number.';
            return null;
        }
        return parseInt(val);
    }
    function setOutput(text) {
        document.getElementById('output').textContent = text;
    }
    function checkEvenOdd() {
        const n = getInput();
        if (n === null) return;
        setOutput(n + ' is ' + (n % 2 === 0 ? 'Even' : 'Odd'));
    }
    function checkPrime() {
        const n = getInput();
        if (n === null) return;
        if (n < 2) { setOutput(n + ' is Not Prime'); return; }
        for (let i = 2; i <= Math.sqrt(n); i++) {
            if (n % i === 0) { setOutput(n + ' is Not Prime'); return; }
        }
        setOutput(n + ' is Prime');
    }
    function calcFactorial() {
        const n = getInput();
        if (n === null) return;
        if (n < 0) { setOutput('Factorial not defined for negative
numbers'); return; }
        let result = 1;
        for (let i = 2; i <= n; i++) result *= i;
        setOutput(n + ' ! = ' + result);
    }
    function checkArmstrong() {
        const n = getInput();
        if (n === null) return;

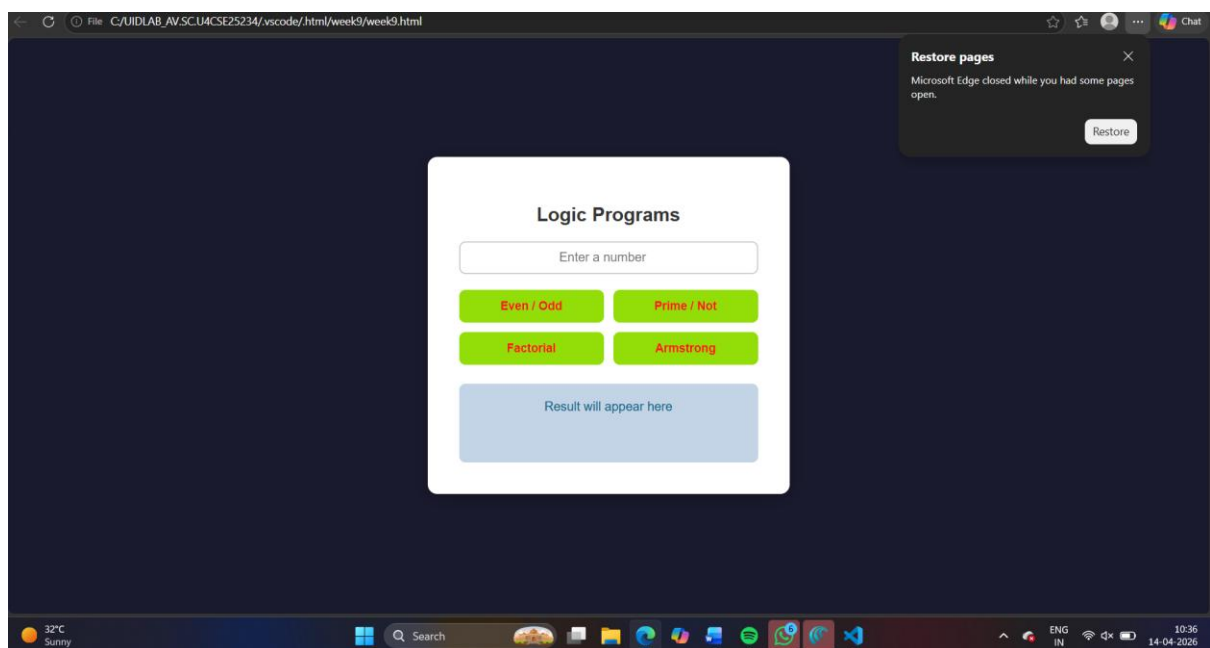
```

```

        const digits = String(n).split('');
        const power  = digits.length;
        const sum     = digits.reduce((acc, d) => acc + Math.pow(Number(d),
power), 0);
        setOutput(n + (sum === n ? ' is an Armstrong number' : ' is Not an
Armstrong number'));
    }
</script>
</body>
</html>

```

OUTPUT:



EXPLANATION:

<DOCTYPE html> : the <!DOCTYPE html> element tells the web browser that this an HTML document

or this is the latest version of HTML

<html> : the <html> element is the root element of an HTML web page

<head> : the <head> element will contain meta data about the HTML web page

<title> : the <title> element specifies a title for the HTML Page

<body> : the <body> element contains all visible content all visible content like text , image , links ,

tables , etc

<h> : we used <h> element for heading

<div>:we use <div> element for encapsulating the divisions and easy to modify

<style> : the <style> element is used to write CSS (designing part) it controls colors, layout, spacing, etc

body : the body selector is used to design the whole page here it sets background color and centers the

content

.container : used to design the main box (card)gives white background, padding, shadow and rounded

corners

input : used to design the input box width, padding, border and text alignment are set

button : used to design buttons gives color, size, border and hover effect

.output : used to design output box where result will be displayed

<input> : the <input> element is used to take user input type="number" allows only numbers id is used to

access it in JavaScript

<div class="buttons"> : groups all buttons together

<button> : used to perform actions when clicked onclick calls JavaScript functions

<div id="output"> : used to display result initially shows "Result will appear here"

<script> : the <script> element is used to write JavaScript code it adds logic and functionality to the

webpage

getElementById() : used to access HTML elements using id

.value : used to get value from input box

.textContent : used to display or change text inside an element

function : used to define a block of code that performs a task

checkEvenOdd() : checks whether number is even or odd uses $n \% 2$

checkPrime() : checks whether number is prime or not uses loop and division logic

calcFactorial() : calculates factorial of a number multiplies numbers from 1 to n

checkArmstrong() : checks Armstrong number sum of digits power equals number

Math.sqrt() : used to find square root (used in prime logic)

parseInt() : converts string to integer

isNaN() : checks if value is not a number

WEEK-10

10A) Aim: write JavaScript conditional statement to find the sign of product of three numbers using functions. Display an alert box with specified sign. write a java script conditional statements to sort 3 numbers using functions display an alert box to show the result. write a java script conditional statement to find largest of five numbers using function and onclick event .display an alert box to show the result .write a java script for loop that will iterate from 0 to 15. For each iteration. it will check if the current number odd or even and display a message on screen. write a java script program to read 6 subjects marks from the student then computes the average marks of students then , this average is used to determine the corresponding grade.

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
  <title>Logic Programs</title>
  <style>
    body {
      font-family: Arial;
      display: flex;
      height: 100vh;
    }
    .left {
      width: 30%;
      background: rgb(99, 99, 12);
      padding: 20px;
      border-right: 3px solid #ccc;
    }
    .right {
      width: 70%;
      background: #f5f5dc;
      padding: 20px;
    }
    input {
      width: 95%;
      padding: 10px;
      margin-bottom: 10px;
    }
    button {
      width: 100%;
      padding: 10px;
      margin-bottom: 10px;
      background: #cfa36b;
    }
  </style>
</head>
</html>
```

```

        border: none;
        cursor: pointer;
    }
    #output {
        background: #e1ce83;
        padding: 20px;
        min-height: 200px;
        white-space: pre-line;
    }
</style>
</head>
<body>
<div class="left">
    <h3>Enter Input</h3>
    <input type="text" id="numbers" placeholder="Click a button to see input
format">
    <button onclick="signProduct()">Sign of Product</button>
    <button onclick="sortNumbers()">Sort Numbers</button>
    <button onclick="largestNumber()">Largest Number</button>
    <button onclick="oddEvenLoop()">Odd / Even 0-15</button>
    <button onclick="studentGrade()">Student Grade</button>
</div>
<div class="right">
    <h3>Output</h3>
    <div id="output">Result will appear here</div>
</div>
<script>
function getNumbers() {
    let input = document.getElementById("numbers").value;
    return input.split(",").map(Number);
}
function signProduct() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
        "Enter 3 numbers like: 0,-1,4";
        return;
    }
    let nums = getNumbers();
    let negative = 0;
    for (let n of nums) {
        if (n < 0) negative++;
    }
}

```

```

    }
    let result = (negative % 2 == 0) ? "+1" : "-1";
    alert(result);
    document.getElementById("output").textContent = "Sign: " + result;
}
function sortNumbers() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
            "Enter 3 numbers like: 0,-1,4";
        return;
    }
    let nums = getNumbers();
    nums.sort((a, b) => b - a);
    alert(nums.join(", "));
    document.getElementById("output").textContent =
        "Sorted: " + nums.join(", ");
}
function largestNumber() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
            "Enter 5 numbers like: -5,-2,-6,0,-1";
        return;
    }
    let nums = getNumbers();
    let max = nums[0];
    for (let n of nums) {
        if (n > max) max = n;
    }
    alert("Largest: " + max);
    document.getElementById("output").textContent =
        "Largest: " + max;
}
function oddEvenLoop() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
            "Enter a number like: 15 (to check from 0 to that number)";
        return;
    }
    let n = Number(input);

    let result = "";
    for (let i = 0; i <= n; i++) {

```

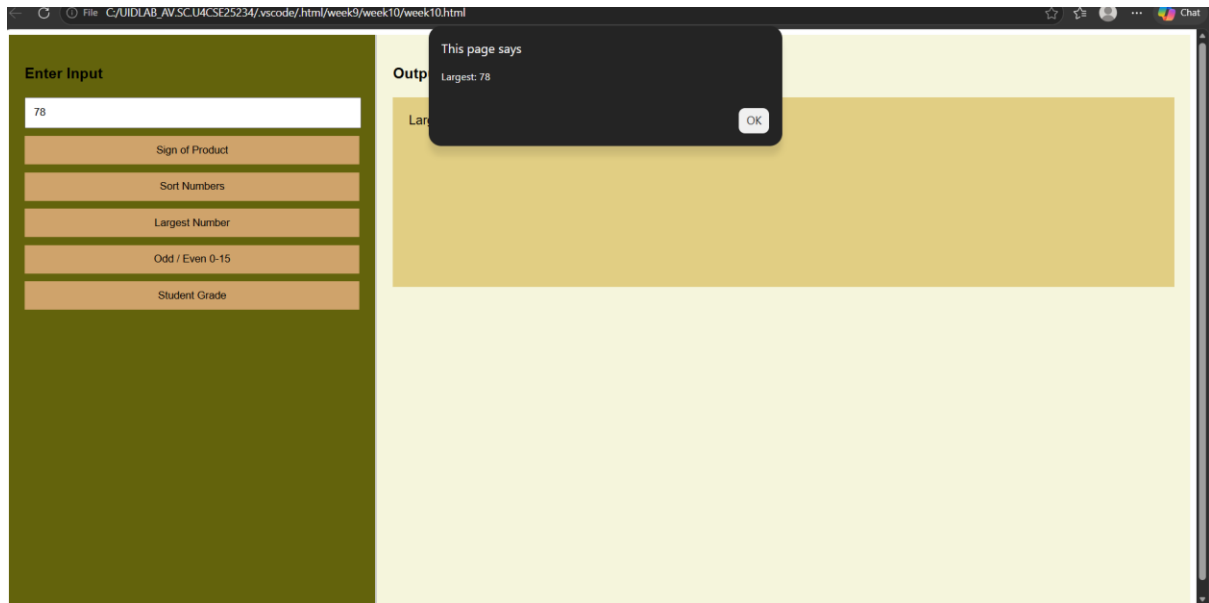
```

        if (i % 2 == 0)
            result += i + " is Even\n";
        else
            result += i + " is Odd\n";
    }

    document.getElementById("output").textContent = result;
}
function studentGrade() {
    let input = document.getElementById("numbers").value;
    if (input === "") {
        document.getElementById("output").textContent =
            "Enter 6 subject marks like: 70,80,90,60,75,85";
        return;
    }
    let nums = getNumbers();
    let sum = 0;
    for (let n of nums) sum += n;
    let avg = sum / nums.length;
    let grade;
    if (avg >= 90) grade = "A";
    else if (avg >= 75) grade = "B";
    else if (avg >= 60) grade = "C";
    else grade = "D";
    alert("Average: " + avg + " Grade: " + grade);
    document.getElementById("output").textContent =
        "Average: " + avg + " | Grade: " + grade;
}
</script>
</body>
</html>

```

OUTPUT:



EXPLANATION:

`<!DOCTYPE html>` : the `<!DOCTYPE html>` element tells the browser that this is an HTML5 document

`<html>` : the `<html>` element is the root element of the web page all elements are written inside this

`<head>` : the `<head>` element contains meta information like title and styles

`<title>` : the `<title>` element specifies the name of the web page it appears on the browser tab

`<style>` : the `<style>` element is used to write CSS for designing the page

`body` : the `body` selector is used to style the whole page here `flex` is used to divide page into two sections

`.left` : this class is used for left side panel contains input box and buttons

`.right` : this class is used for right side panel used to display output

`input` : the `<input>` element is used to take user input here `type="text"` is used to enter numbers like 1,2,3

button : the <button> element is used to perform actions onclick is used to call JavaScript functions

<div> : the <div> element is used to group elements helps in layout design

#output : this is an id used to display result .textContent is used to change its text

<script> : the <script> element is used to write JavaScript code

function : used to define a block of code that performs a task

getElementById() : used to access HTML elements using id

.value : used to get value from input box

.split(",") : used to convert input string into array using comma

.map(Number) : converts string values into numbers

for loop : used to repeat code multiple times

if condition : used to check conditions

alert() : used to display popup message

sort() : used to sort numbers in array

max logic : used to find largest number using loop

oddEvenLoop() : checks numbers from 0 to given input tells whether each number is even or odd

studentGrade() : calculates average of marks assigns grade based on conditions

.textContent : used to display output on the web page

WEEK~11.

11a) A website requires users to enter a username. The system should display a message while typing if the username is less than 5 characters.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Username Validation</title>
</head>
<body>

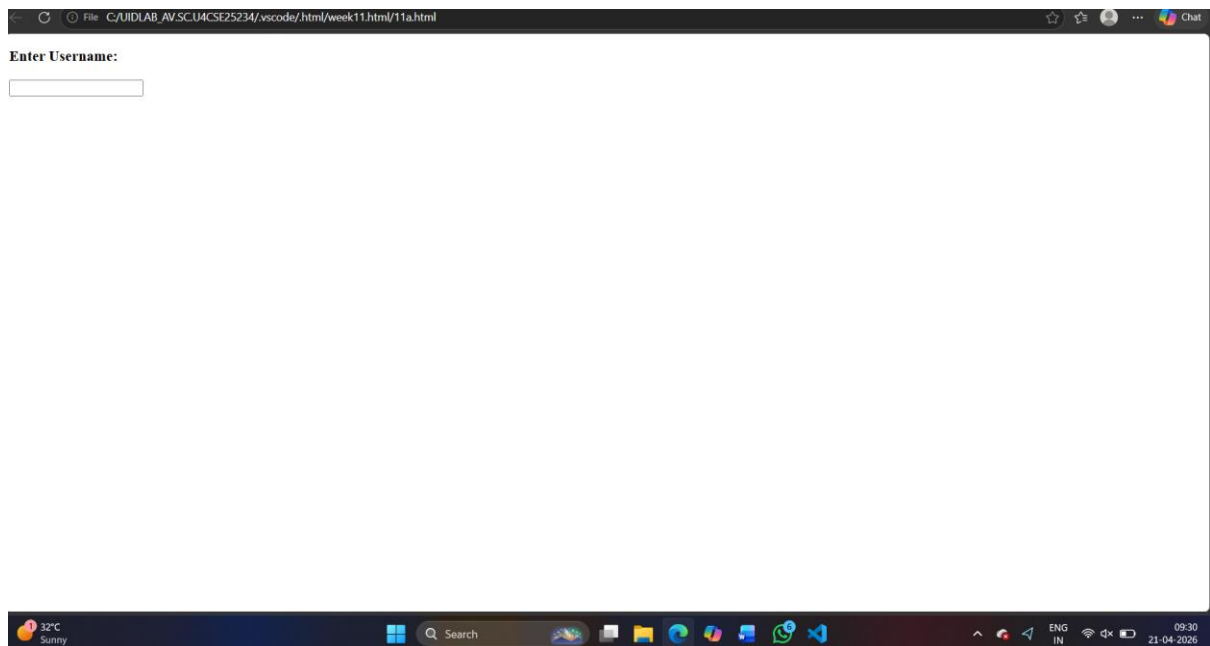
<h3>Enter Username:</h3>
<input type="text" id="username" onkeyup="checkUsername()">
<p id="msg"></p>

<script>
function checkUsername() {
  let username = document.getElementById("username").value;
  let msg = document.getElementById("msg");

  if (username.length < 5) {
    msg.innerHTML = "Username must be at least 5 characters";
    msg.style.color = "red";
  } else {
    msg.innerHTML = "Valid username";
    msg.style.color = "green";
  }
}
</script>

</body>
</html>
```

Output:



Explanation:

1. `document.getElementById(...).value`: This reaches into the HTML, finds the input box by its ID, and pulls out whatever the user typed.

2. `innerHTML`: Changes the actual text inside the `<p>` tag.

3. `<p id="msg"></p>`: An empty paragraph. It acts as a placeholder where your JavaScript will inject error messages like "Minimum 5 letters required".

4. `<input type="text">`: Creates a standard text box for user input.

5. `id="username"`: This is a unique identifier. It acts like a "hook" that allows JavaScript to find this specific input box among all other elements on the page.

6. `onkeyup="checkUsername()"`: Every time the user releases a key (after pressing it), the `checkusername()` function is triggered.

7. `length`: This property counts the number of characters in the string.

11b) A registration form should validate email format before submission. If invalid, the form should not be submitted.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Email Validation</title>
</head>
<body>

<form onsubmit="return validateEmail()">
  Email: <input type="text" id="email">
  <input type="submit" value="Submit">
</form>

<p id="msg"></p>

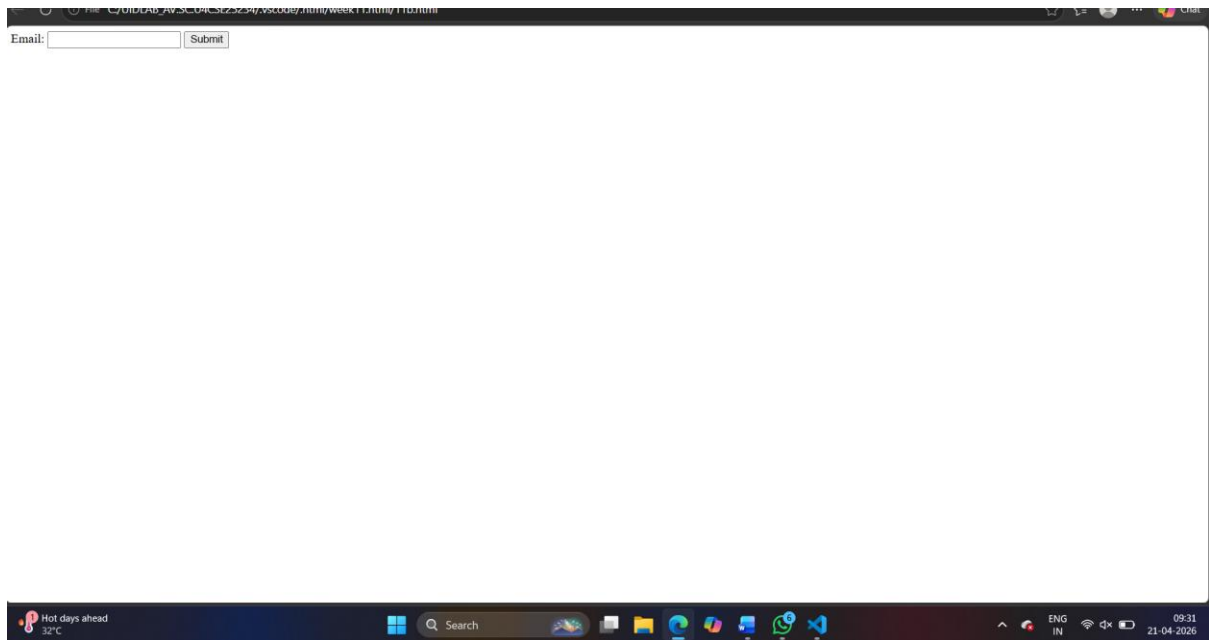
<script>
function validateEmail() {
  let email = document.getElementById("email").value;
  let msg = document.getElementById("msg");

  let pattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

  if (!email.match(pattern)) {
    msg.innerHTML = "Invalid Email Format";
    msg.style.color = "red";
    return false; // stop submission
  }
  return true;
}
</script>

</body>
</html>
```

Output:



Explanation:

1. `document.getElementById(...).value`: This reaches into the HTML, finds the input box by its ID, and pulls out whatever the user typed.
2. `.innerHTML`: Changes the actual text inside the `<p>` tag.
3. `<p id="msg"></p>`: An empty paragraph. It acts as a placeholder where your JavaScript will inject error messages like "Invalid Email."
4. `<input type="text" id="mail" placeholder="Enter email">`: The placeholder attribute provides a faint hint inside the box that disappears once the user starts typing.
5. `^[a-zA-Z0-9._%+-]+`: Matches the username part before the `@`. It allows letters, numbers, dots, underscores, and plus signs.
6. `@[a-zA-Z0-9.-]+`: Matches the domain name.

7. `\.[a-zA-Z]{2,4}$`: Matches the extension (like .com or .in). The {2,4} means the extension must be between 2 and 4 characters long.

8. `document.getElementById("msg").style.color` : It gives the user immediate visual feedback: Red for a mistake and Green for success.

11c) A system should check password strength while typing and display whether it is weak or strong.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Password Strength</title>
</head>
<body>

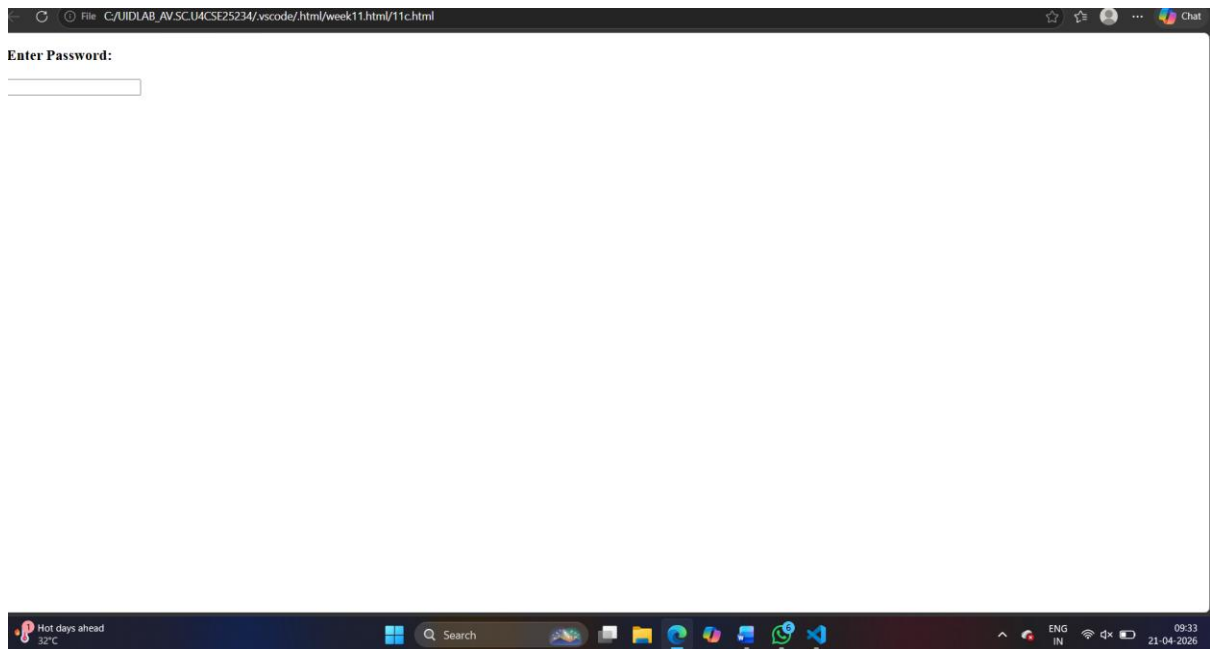
<h3>Enter Password:</h3>
<input type="password" id="password" onkeyup="checkPassword()">
<p id="msg"></p>

<script>
function checkPassword() {
  let pass = document.getElementById("password").value;
  let msg = document.getElementById("msg");

  if (pass.length < 6) {
    msg.innerHTML = "Weak Password";
    msg.style.color = "red";
  } else {
    msg.innerHTML = "Strong Password";
    msg.style.color = "green";
  }
}
</script>

</body>
</html>
```

Output:



Explanation:

1. `document.getElementById("...").value`: This reaches into the HTML, finds the input box by its ID, and pulls out whatever the user typed.
2. `.innerHTML`: Changes the actual text inside the `<p>` tag.
3. `onkeyup`: Fires when the user releases the key.
4. `.length`: This is a built-in JavaScript property that counts how many characters are in the string.
5. `document.getElementById("msg").style.color`: It gives the user immediate visual feedback: Red for a mistake and Green for success.
6. `<input type="password">`: Creates a text box where characters are masked.

11D) A form should validate a mobile number when the user leaves the input field. The number must be exactly 10 digits.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Mobile Validation</title>
</head>
<body>

<h3>Enter Mobile Number:</h3>
<input type="text" id="mobile" onblur="validateMobile()">
<p id="msg"></p>

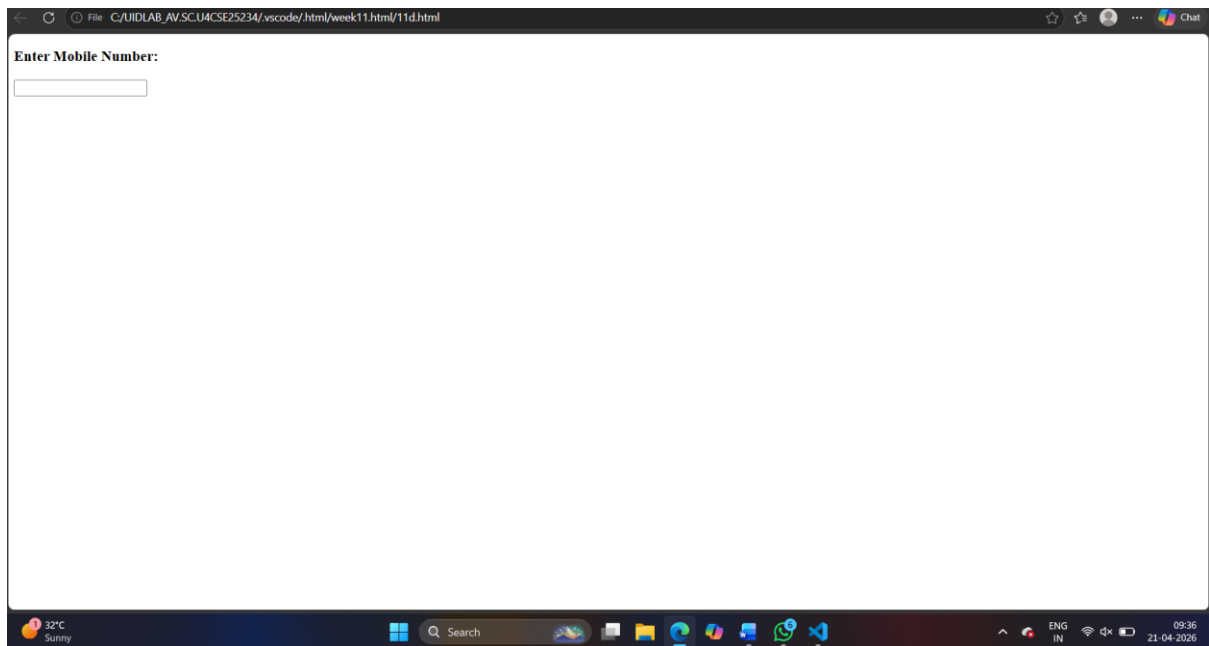
<script>
function validateMobile() {
  let mobile = document.getElementById("mobile").value;
  let msg = document.getElementById("msg");

  let pattern = /^[0-9]{10}$/;

  if (!mobile.match(pattern)) {
    msg.innerHTML = "Enter a valid 10-digit mobile number";
    msg.style.color = "red";
  } else {
    msg.innerHTML = "Valid mobile number";
    msg.style.color = "green";
  }
}
</script>

</body>
</html>
```

Output:



Explanation:

1. `document.getElementById("."").value`: This reaches into the HTML, finds the input box by its ID, and pulls out whatever the user typed.

2. `.innerHTML`: Changes the actual text inside the `<p>` tag.

3. `let pattern = /^[0-9]{10}$/;` : This is a regular expression.

· `^` : Start of the string.

· `[0-9]` : Only digits are allowed.

· `{10}` : Exactly 10 digits required.

· `$` : End of the string.

4. `document.getElementById("msg").style.color` : It gives the user immediate visual feedback: Red for a mistake and Green for success.
5. `oninput="Mobilenumber()"`:fires every single time the value inside the box changes.
6. `.match(pattern)` : Compares the user's input against the Regex.

11E) Design an HTML page, such that the registration form must validate:

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Form</title>
</head>
<body>

<h2>Registration Form</h2>

<form onsubmit="return validateForm()">
  Name: <input type="text" id="name"><br><br>
  Email: <input type="text" id="email"><br><br>
  Password: <input type="password" id="password"><br><br>
  <input type="submit" value="Register">
</form>
```

```
<p id="msg"></p>

<script>
function validateForm() {
    let name = document.getElementById("name").value;
    let email = document.getElementById("email").value;
    let password = document.getElementById("password").value;
    let msg = document.getElementById("msg");

    let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

    if (name === "") {
        msg.innerHTML = "Name cannot be empty";
        return false;
    }

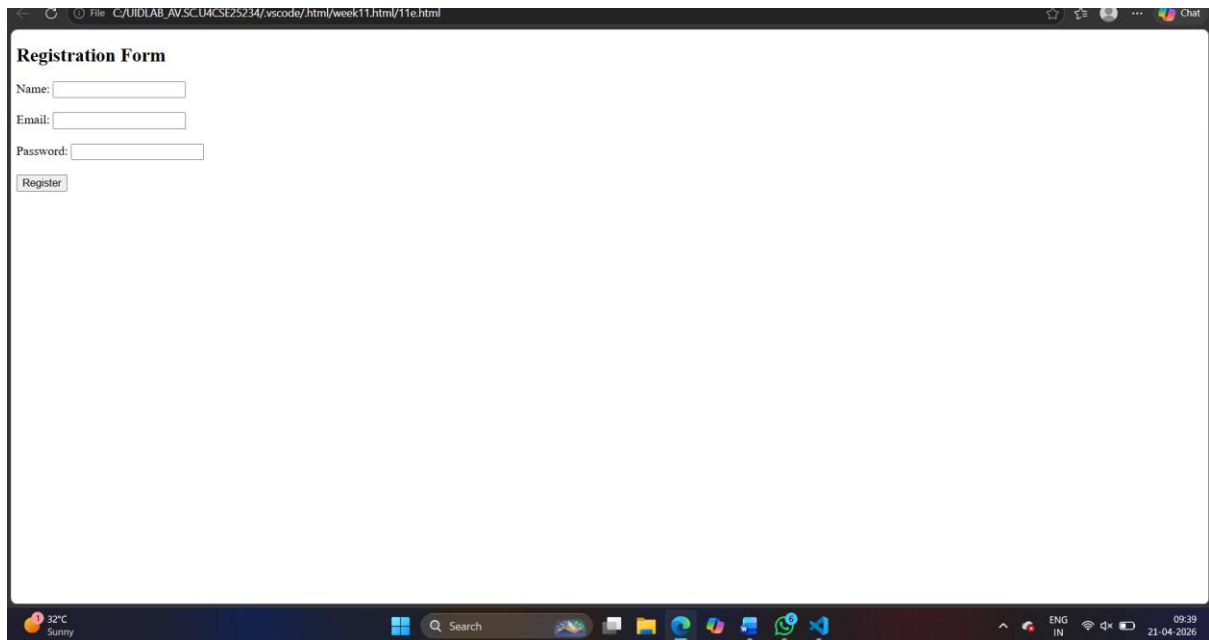
    if (!email.match(emailPattern)) {
        msg.innerHTML = "Invalid Email";
        return false;
    }

    if (password.length < 6) {
        msg.innerHTML = "Password must be at least 6 characters";
        return false;
    }

    return true;
}
</script>

</body>
</html>
```


Output:



The screenshot shows a web browser window with a registration form. The browser's address bar displays the file path: `C:\UIDLAB_AVSC\U4CSE25234\vscode\html\week11.html\11e.html`. The form is titled "Registration Form" and contains three input fields: "Name:", "Email:", and "Password:". Below these fields is a "Register" button. The browser's status bar at the bottom shows the system tray with a weather widget (32°C Sunny), a search bar, and various system icons including language (ENG IN), network, and battery status. The time displayed is 09:39 on 21-04-2026.

Explanation:

1. `document.getElementById(...).value`: This reaches into the HTML, finds the input box by its ID, and pulls out whatever the user typed.

2. `return false`: Returning false stops the form from submitting.

3. `return true`: Returning true means the data is valid and the form can proceed.

4. `<form>`: Defines a container for user input.

5. `<input>`:

- `type="text"`: Creates a basic single-line text box.

- `type="password"`: Creates a text box where characters are masked.
- `type="submit"`: Creates a button that automatically triggers the form's submission.
- `id="..."`: A unique identifier for the element, which the JavaScript uses to "grab" the data entered by the user.

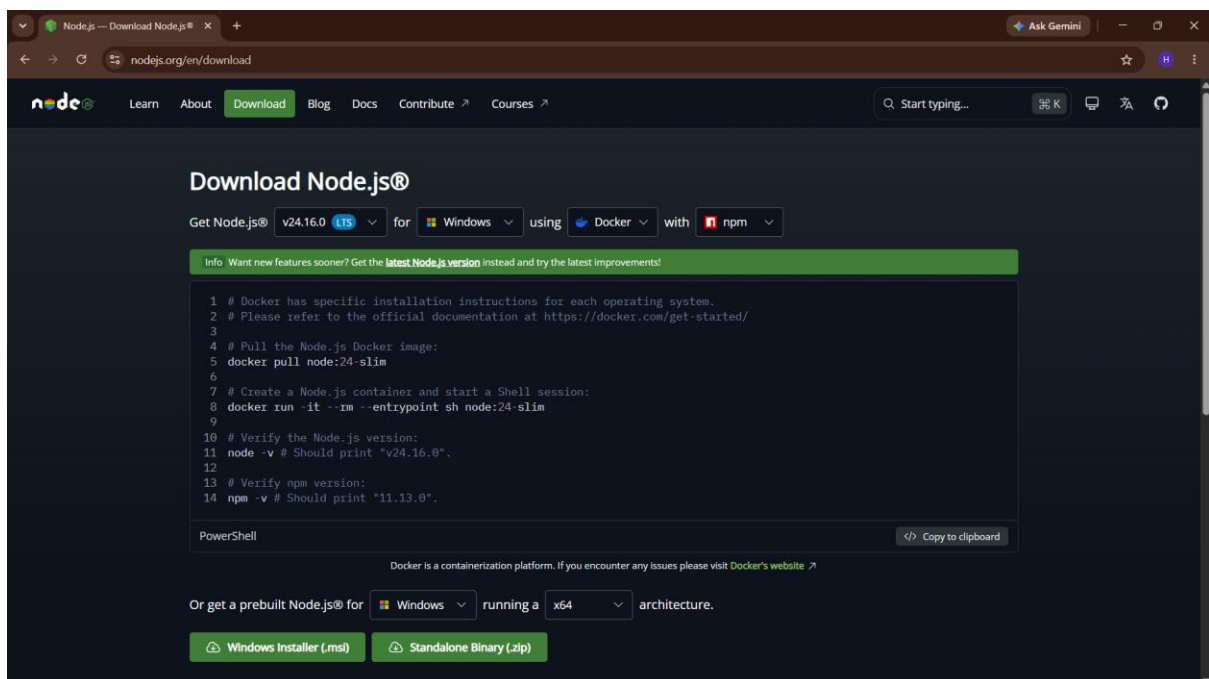
6. `<p id="msg"></p>`: An empty paragraph. It acts as a placeholder where your JavaScript will inject error messages like "Invalid Email."

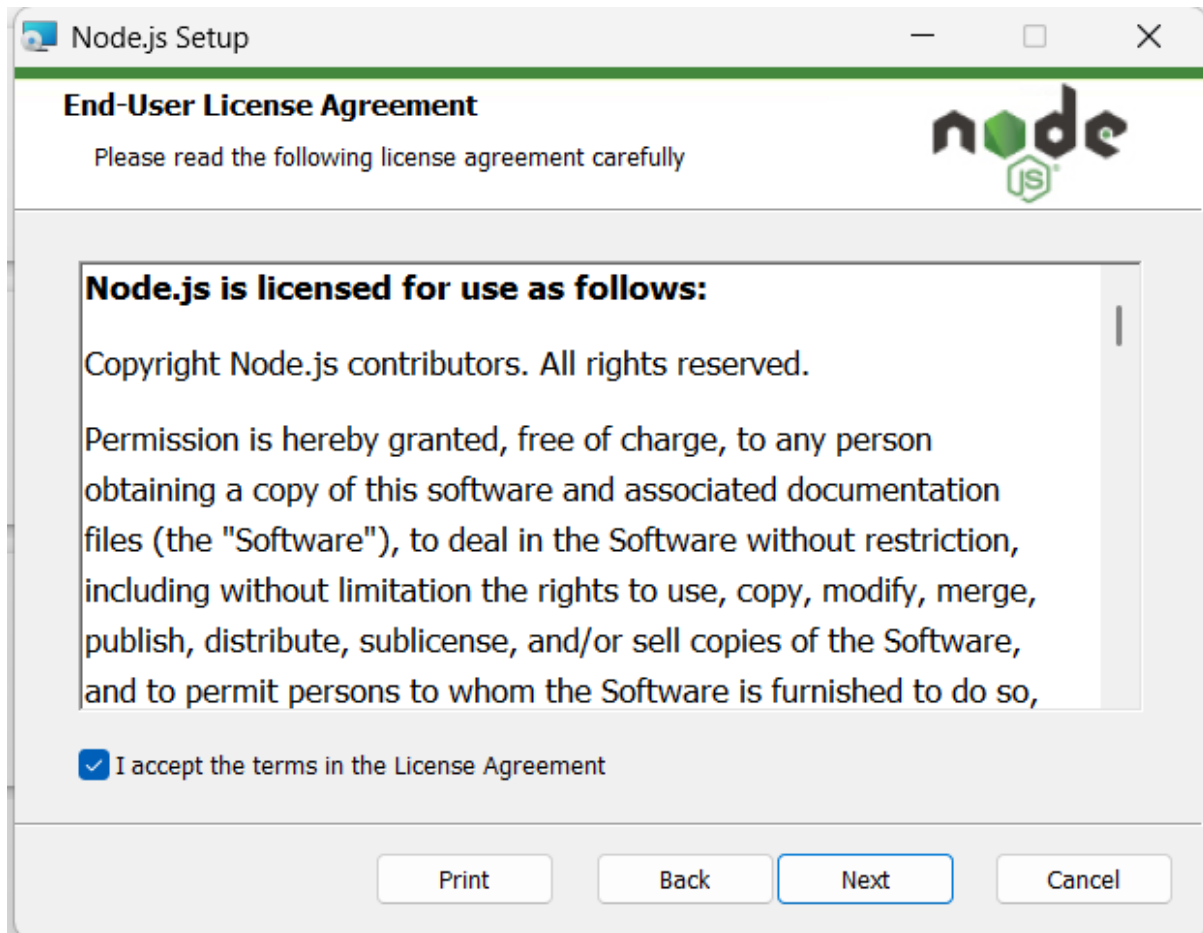
WEEK-12

Aim: Aim: Installation of electron

Steps for installation of electron:

Step1: Open Google Chrome browser.

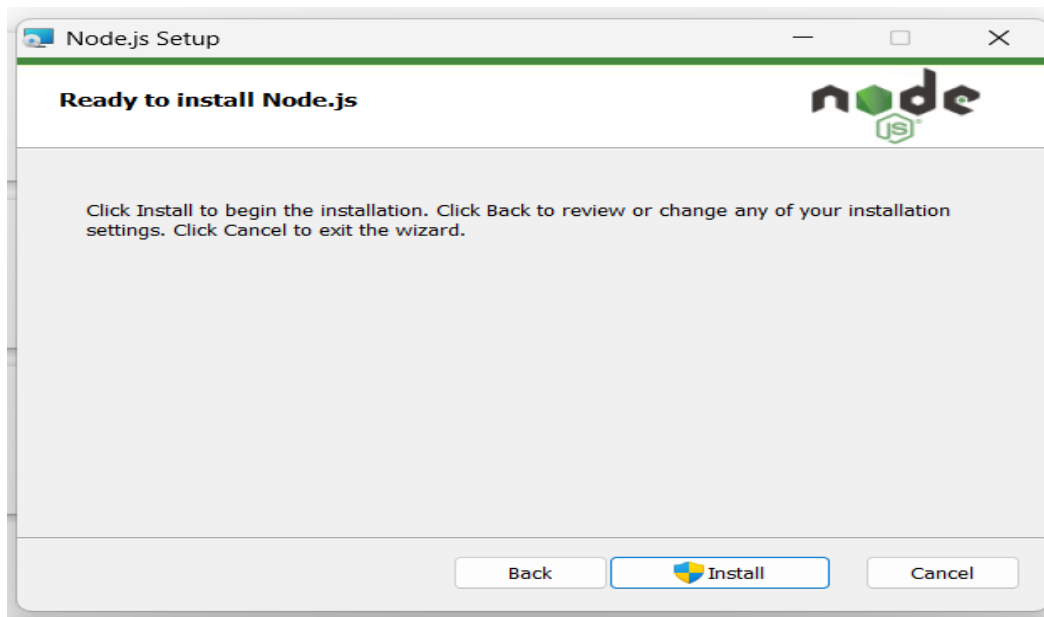




Node.js automatically installs:

- Node

npm (Node Package manager)



Step2:

Open VS Code.

Open terminal:

```
PS C:\Users\srishanth\OneDrive\Documents\node js> npm -v
11.13.0
PS C:\Users\srishanth\OneDrive\Documents\node js> node
Welcome to Node.js v24.16.0.
Type ".help" for more information.
>
```

Step3:

Create New Empty Folder

- Create a new folder anywhere

Example:

ElectronApp

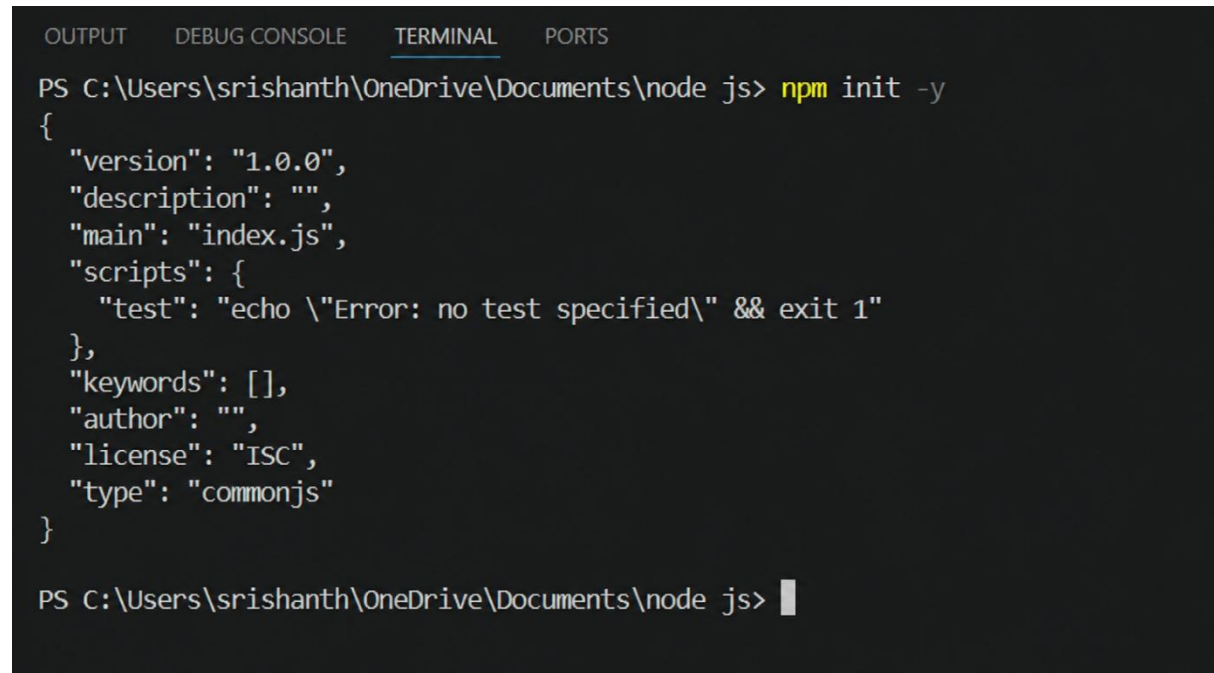
Step4 — Initialize Node Project

Run:

```
npm init -y
```

This creates:

package.json

A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal shows the command 'PS C:\Users\srishanth\OneDrive\Documents\node js> npm init -y' followed by the output of a JSON object: { "version": "1.0.0", "description": "", "main": "index.js", "scripts": { "test": "echo \"Error: no test specified\" && exit 1" }, "keywords": [], "author": "", "license": "ISC", "type": "commonjs" }. The prompt 'PS C:\Users\srishanth\OneDrive\Documents\node js>' is shown again at the bottom with a cursor.

```
PS C:\Users\srishanth\OneDrive\Documents\node js> npm init -y
{
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}

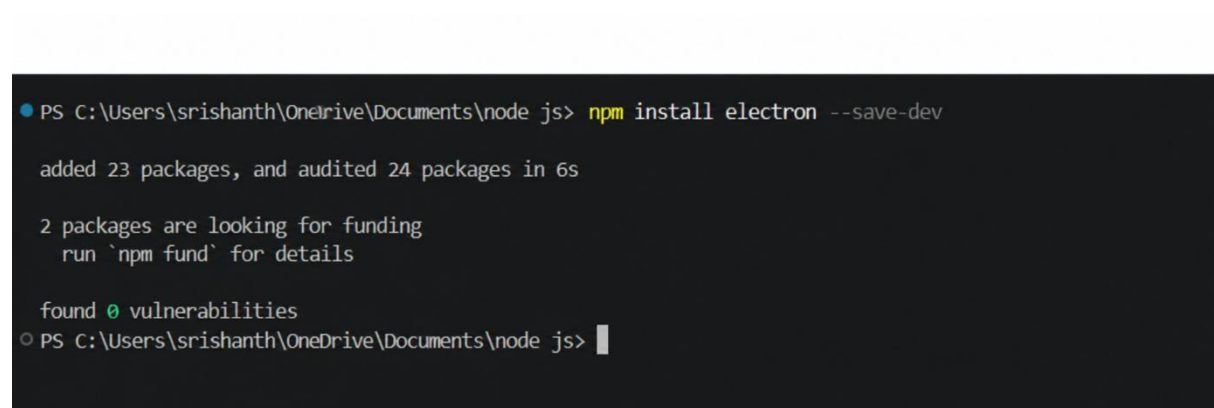
PS C:\Users\srishanth\OneDrive\Documents\node js> █
```

Step5 — Install Electron

Run:

```
npm install electron --save-dev
```

Wait until installation finishes

A screenshot of a terminal window with a dark background. It shows the command 'PS C:\Users\srishanth\OneDrive\Documents\node js> npm install electron --save-dev' followed by the output: 'added 23 packages, and audited 24 packages in 6s', '2 packages are looking for funding', 'run `npm fund` for details', and 'found 0 vulnerabilities'. The prompt 'PS C:\Users\srishanth\OneDrive\Documents\node js>' is shown at the bottom with a cursor.

```
PS C:\Users\srishanth\OneDrive\Documents\node js> npm install electron --save-dev
added 23 packages, and audited 24 packages in 6s

2 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\srishanth\OneDrive\Documents\node js> █
```

Step6 — Modify package.json

Open package.json

Add:

"main": "main.js",

Inside "scripts" add:

"start": "electron ."

Final example:

```
{  
  "name": "electronapp",  
  "version": "1.0.0",  
  "main": "main.js",  
  "scripts": {  
    "start": "electron ."  
  }  
}
```

Step7 — Open Electron Official Website

Open Chrome.

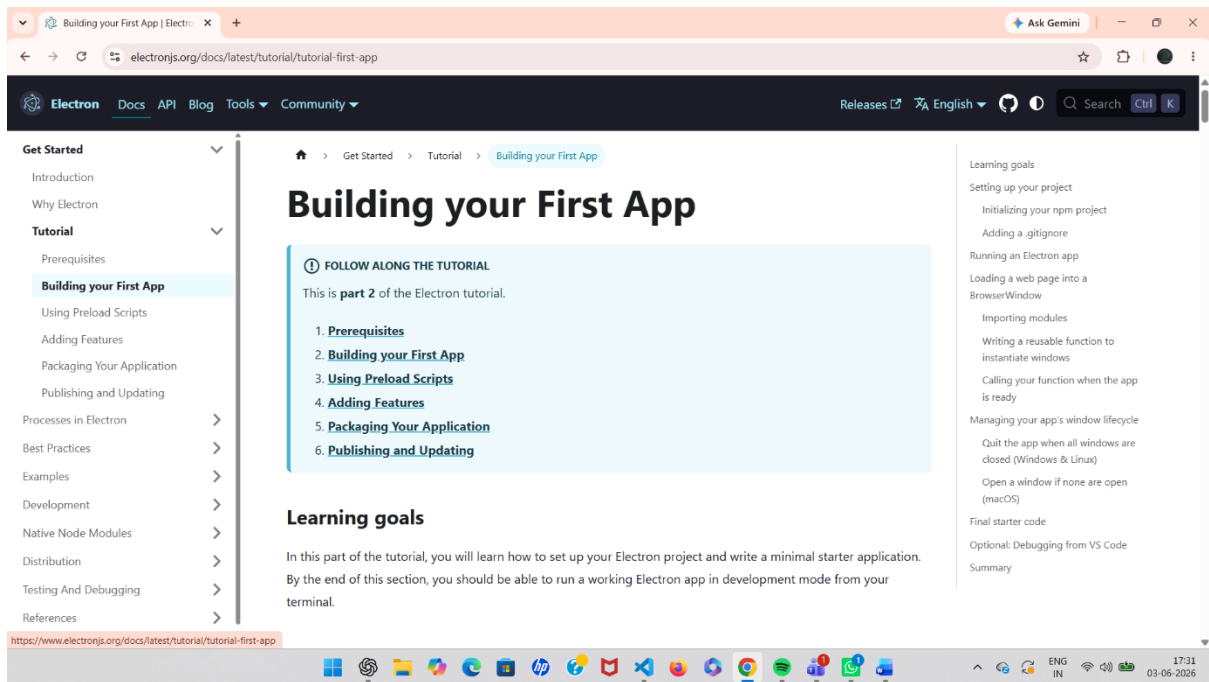
Go to:

[Electron.js Official Website](https://electronjs.org/)

Navigate:

Docs → Tutorial → Building Your First App

Copy the **Final Starter Code**

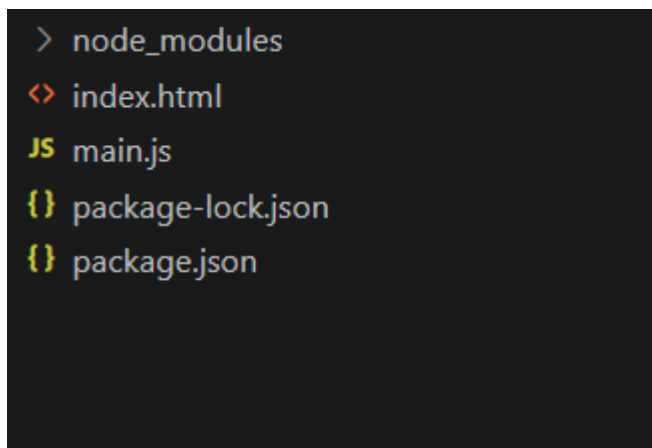


Step8 — Create Required Files

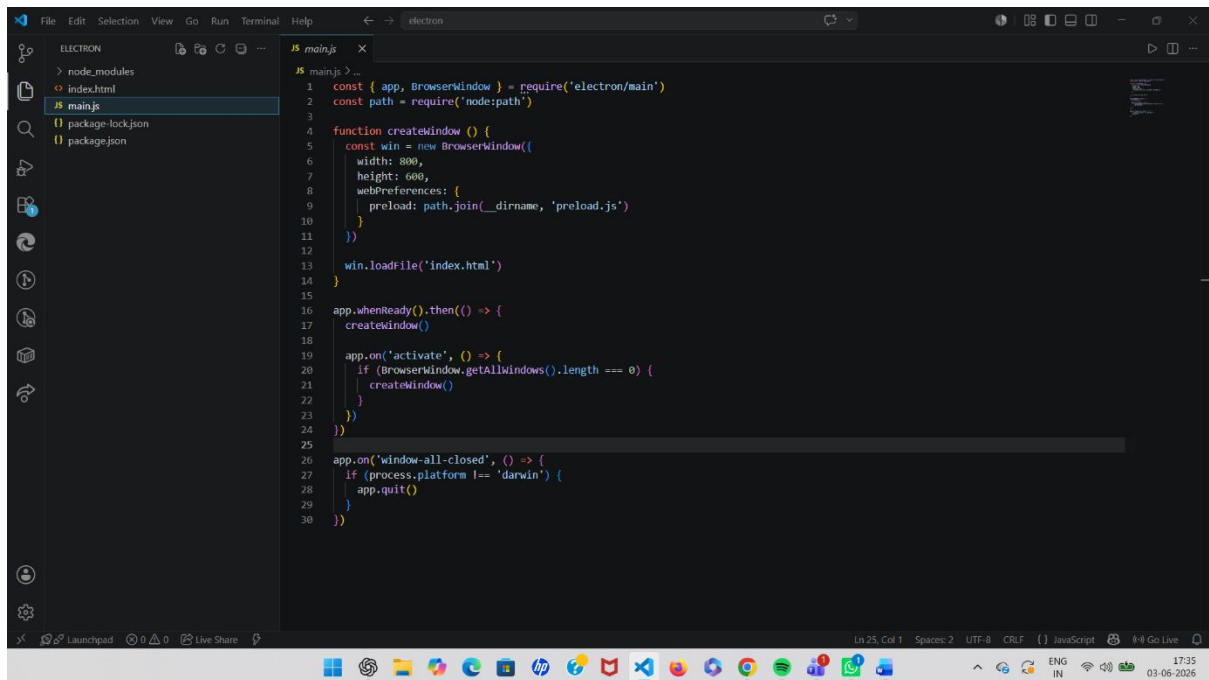
Inside your project folder create:

main.js

index.html

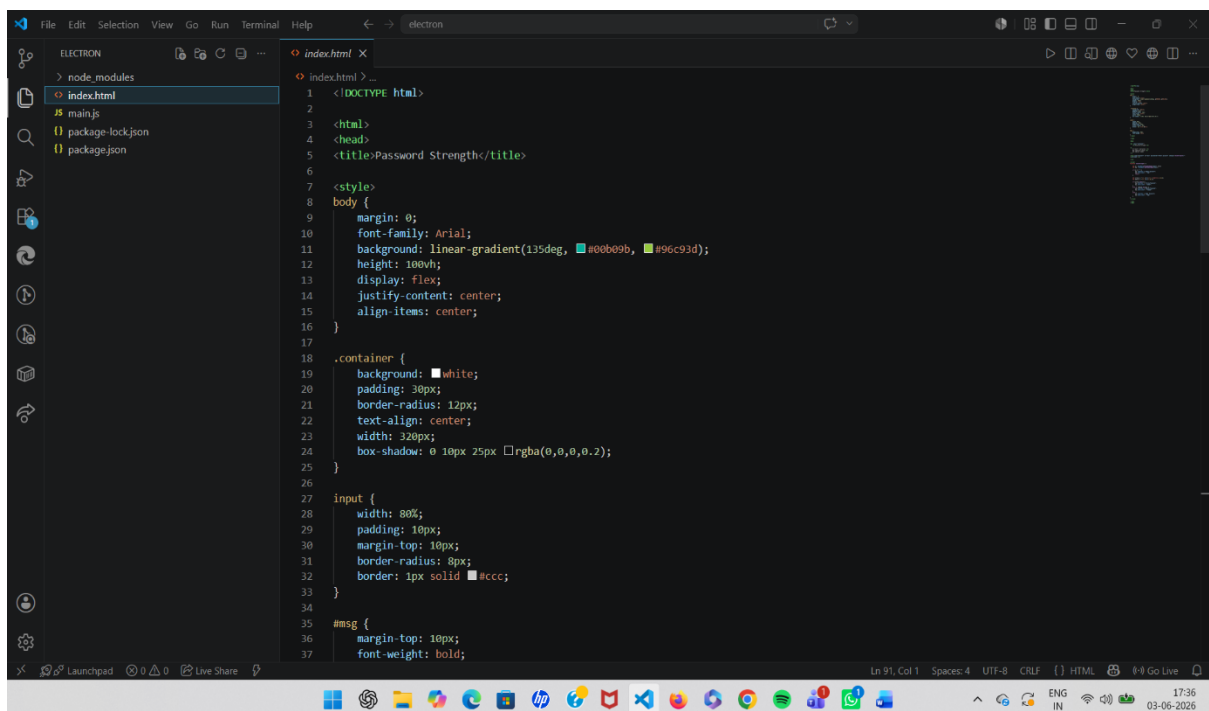


Step9 — Paste Electron Code in main.js



```
1 const { app, BrowserWindow } = require('electron/main')
2 const path = require('node:path')
3
4 function createWindow () {
5   const win = new BrowserWindow({
6     width: 800,
7     height: 600,
8     webPreferences: {
9       preload: path.join(__dirname, 'preload.js')
10    }
11  })
12
13  win.loadFile('index.html')
14 }
15
16 app.whenReady().then(() => {
17   createWindow()
18
19   app.on('activate', () => {
20     if (BrowserWindow.getAllWindows().length === 0) {
21       createWindow()
22     }
23   })
24 })
25
26 app.on('window-all-closed', () => {
27   if (process.platform !== 'darwin') {
28     app.quit()
29   }
30 })
```

Step10 — Add HTML File



```
1 <!DOCTYPE html>
2
3 <html>
4 <head>
5   <title>Password Strength</title>
6
7   <style>
8     body {
9       margin: 0;
10      font-family: Arial;
11      background: linear-gradient(135deg, #00b09b, #96c93d);
12      height: 100vh;
13      display: flex;
14      justify-content: center;
15      align-items: center;
16    }
17
18    .container {
19      background: white;
20      padding: 30px;
21      border-radius: 12px;
22      text-align: center;
23      width: 320px;
24      box-shadow: 0 10px 25px rgba(0,0,0,0.2);
25    }
26
27    input {
28      width: 80%;
29      padding: 10px;
30      margin-top: 10px;
31      border-radius: 8px;
32      border: 1px solid #ccc;
33    }
34
35    #msg {
36      margin-top: 10px;
37      font-weight: bold;
```

Step11 — Verify File Names

Make sure:

```
win.loadFile('index.html')
```

matches your actual HTML filename.

Step 12 — Run the Electron App

In terminal run:

```
npm start
```

Your Electron desktop app window will open

Description:

Description of Modules Used

1. Node.js

Node.js is a JavaScript runtime environment used to execute JavaScript code outside the browser. It provides the backend environment required for Electron applications. In this project, Node.js is used to initialize the project, install dependencies, and manage packages using npm.

2. npm (Node Package Manager)

npm is the default package manager for Node.js. It is used to install Electron and other required libraries into the project. Commands such as `npm init -y`, `npm install electron --save-dev`, and `npm start` are executed using npm.

3. Electron.js

Electron.js is an open-source framework used for building desktop applications using HTML, CSS, and JavaScript. It combines Chromium and Node.js, allowing web applications to run as desktop software on Windows, Linux, and macOS.

4. main.js

The main.js file is the main entry point of the Electron application. It creates the application window using the `BrowserWindow` module and controls the lifecycle of the desktop application.

Main functions of main.js:

- Creates the application window
- Loads the HTML file

- Handles application startup and closing events

5. BrowserWindow Module

BrowserWindow is an Electron module used to create and manage desktop windows. It allows the application to display web pages inside a native desktop window.

Features:

- Set window size
- Load HTML pages
- Control window behavior

6. app Module

The app module controls the overall lifecycle of the Electron application. It handles events such as:

- Application startup
- Closing windows
- Activating the application

7. package.json

The package.json file contains project information and configuration details. It stores:

- Project name and version
- Installed dependencies
- Main entry file
- Scripts used to run the application

8. index.html

The index.html file contains the user interface of the Electron application. It is loaded inside the Electron window and displays the frontend content using HTML and CSS.

9. preload.js

The preload.js file acts as a bridge between the renderer process and Node.js environment. It is executed before the web page loads and is commonly used for secure communication between frontend and backend components.

Introduction to Electron.js

Electron.js is an open-source framework used to build cross-platform desktop applications using web technologies such as HTML, CSS, and JavaScript. It was developed and maintained by [Electron.js](#).

Electron combines two major technologies:

- Chromium browser engine for the user interface
- Node.js runtime for backend and system-level operations

This allows developers to create desktop applications using the same technologies used for web development.

Applications built with Electron can run on:

- Windows
- macOS
- Linux

Electron is widely used for developing modern desktop applications because it provides:

- Easy development process
- Cross-platform compatibility

- Access to native system features
- Support for modern web technologies

Popular applications built using Electron include:

- Visual Studio Code
- Discord
- Slack
- WhatsApp Desktop
- Spotify Desktop

In this project, Electron is used to create a desktop application by setting up a Node.js project, installing Electron through npm, and running the application using the Electron runtime.

